

BFS Traversal

Step 1: Start at A

Legend

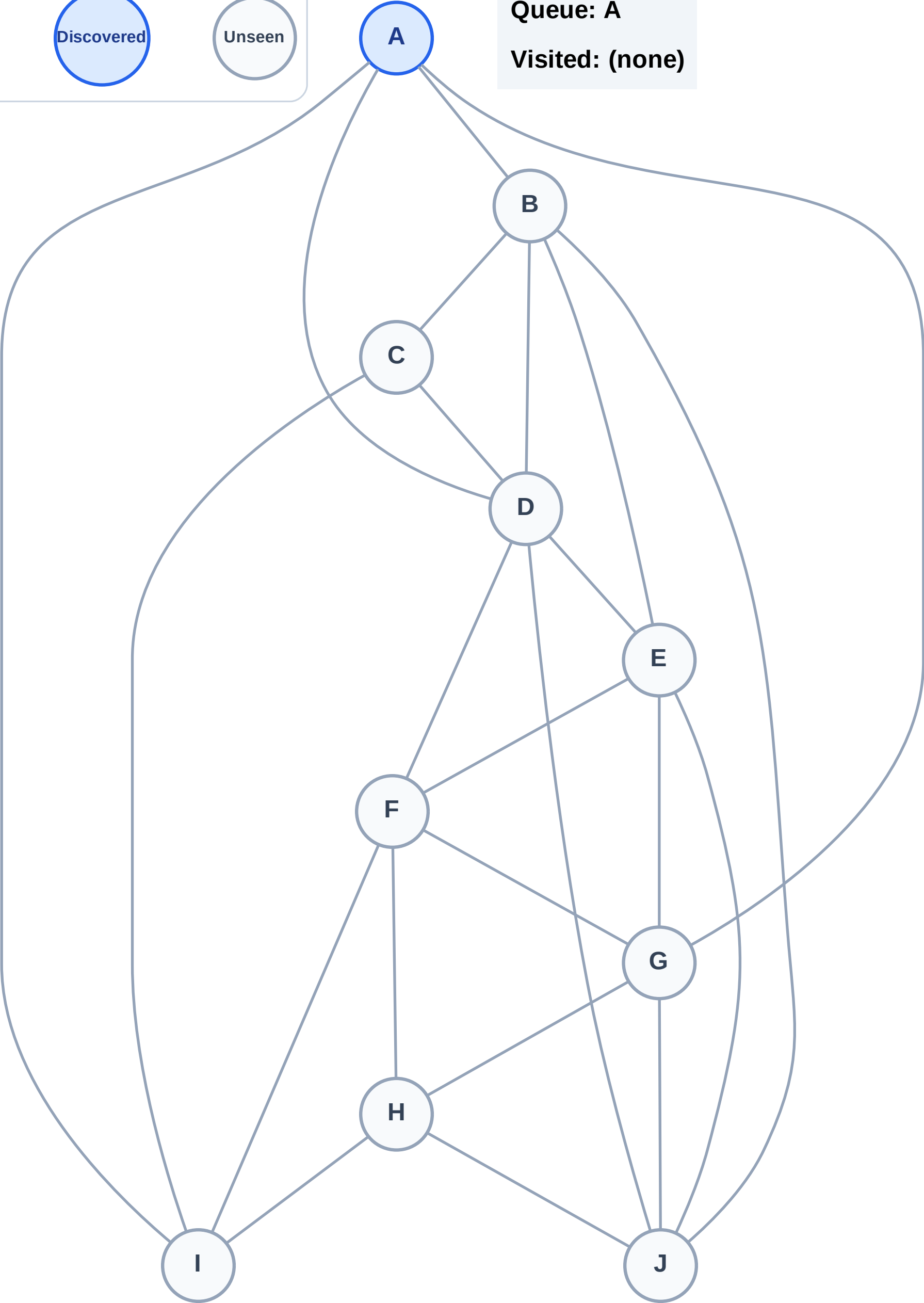
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

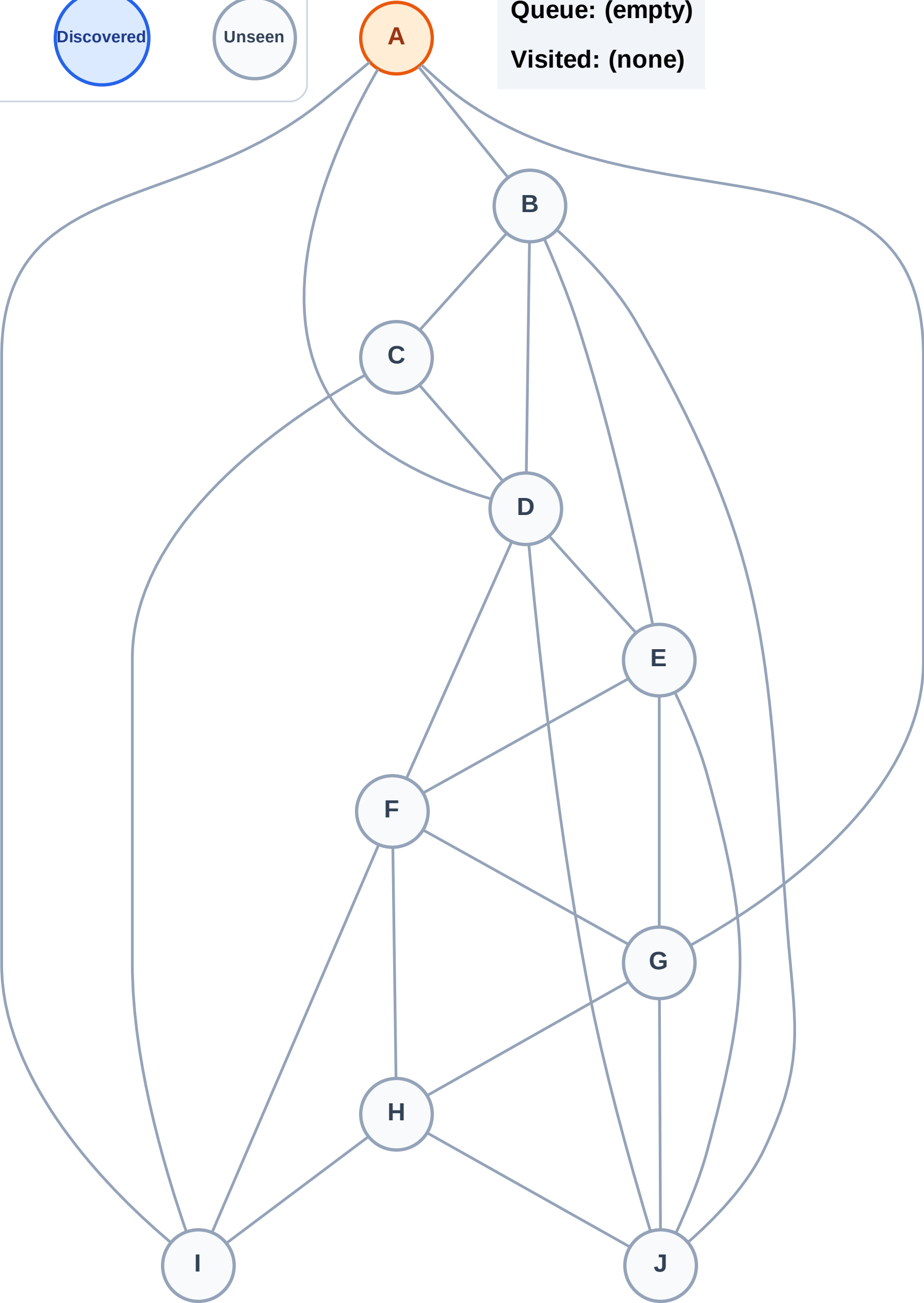
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, G, I from A

Legend

Complete

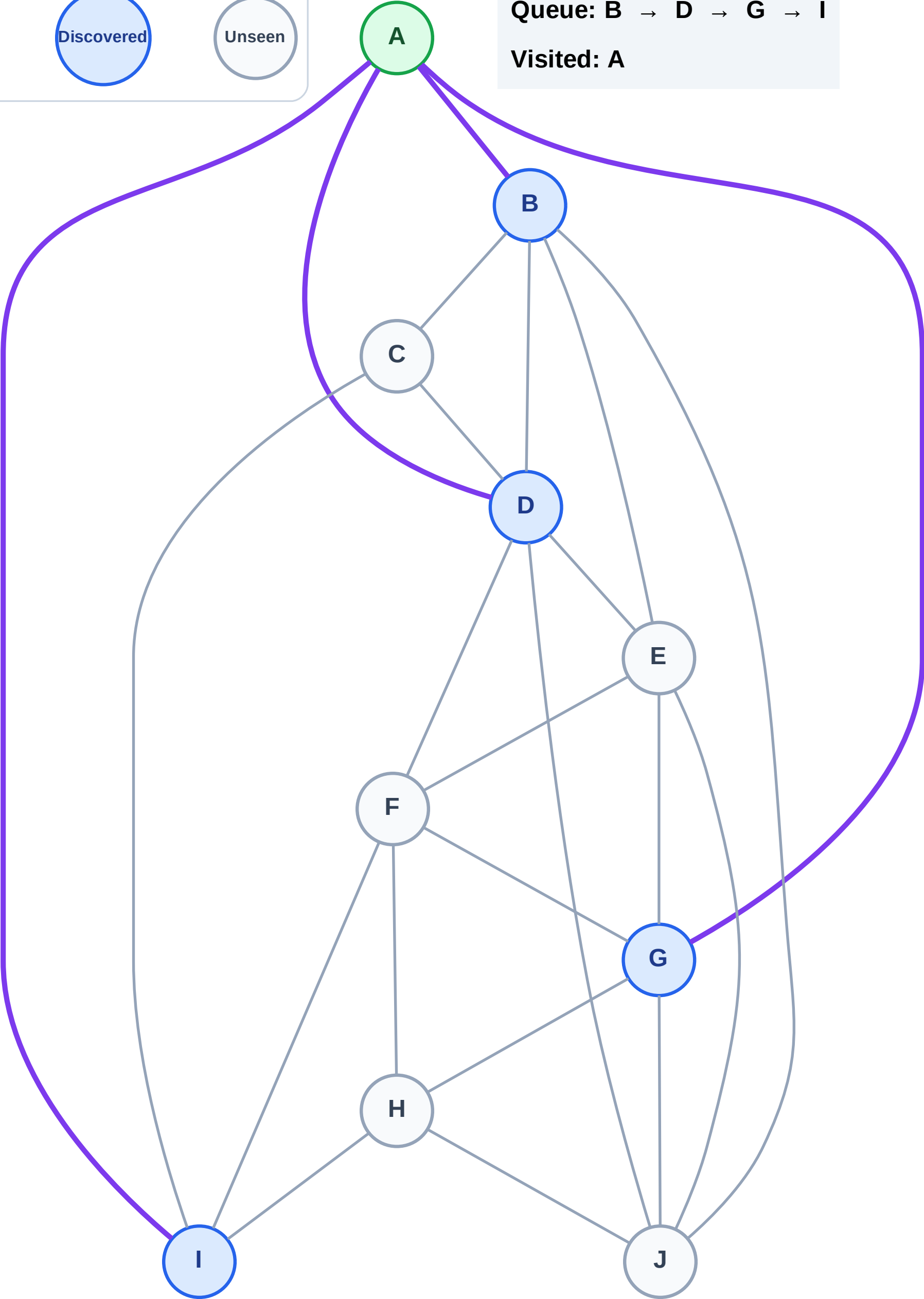
Processing

Discovered

Unseen

Queue: B → D → G → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

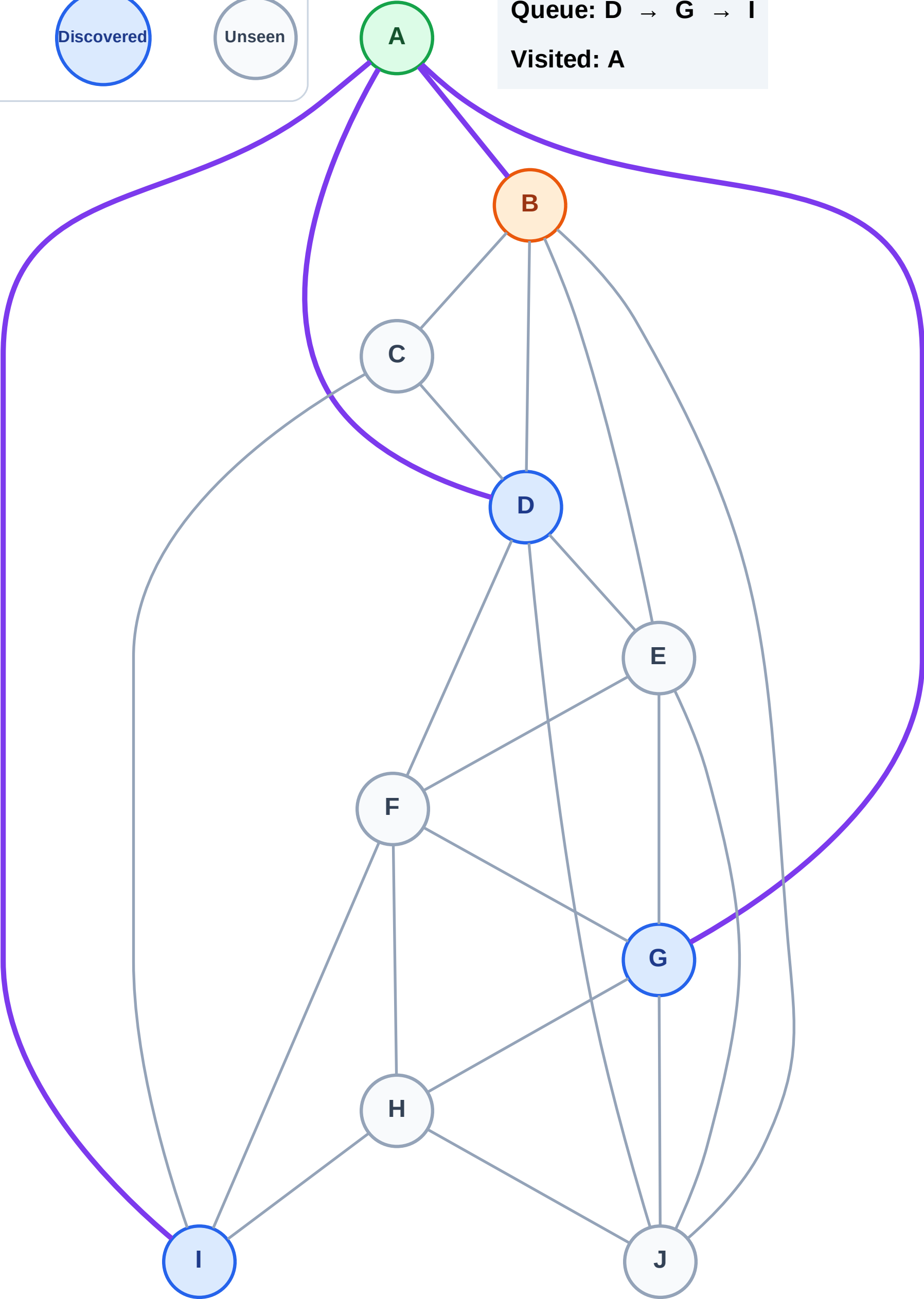
Processing

Discovered

Unseen

Queue: D → G → I

Visited: A



BFS Traversal

Step 5: Discover C, E, J from B

Legend

Complete

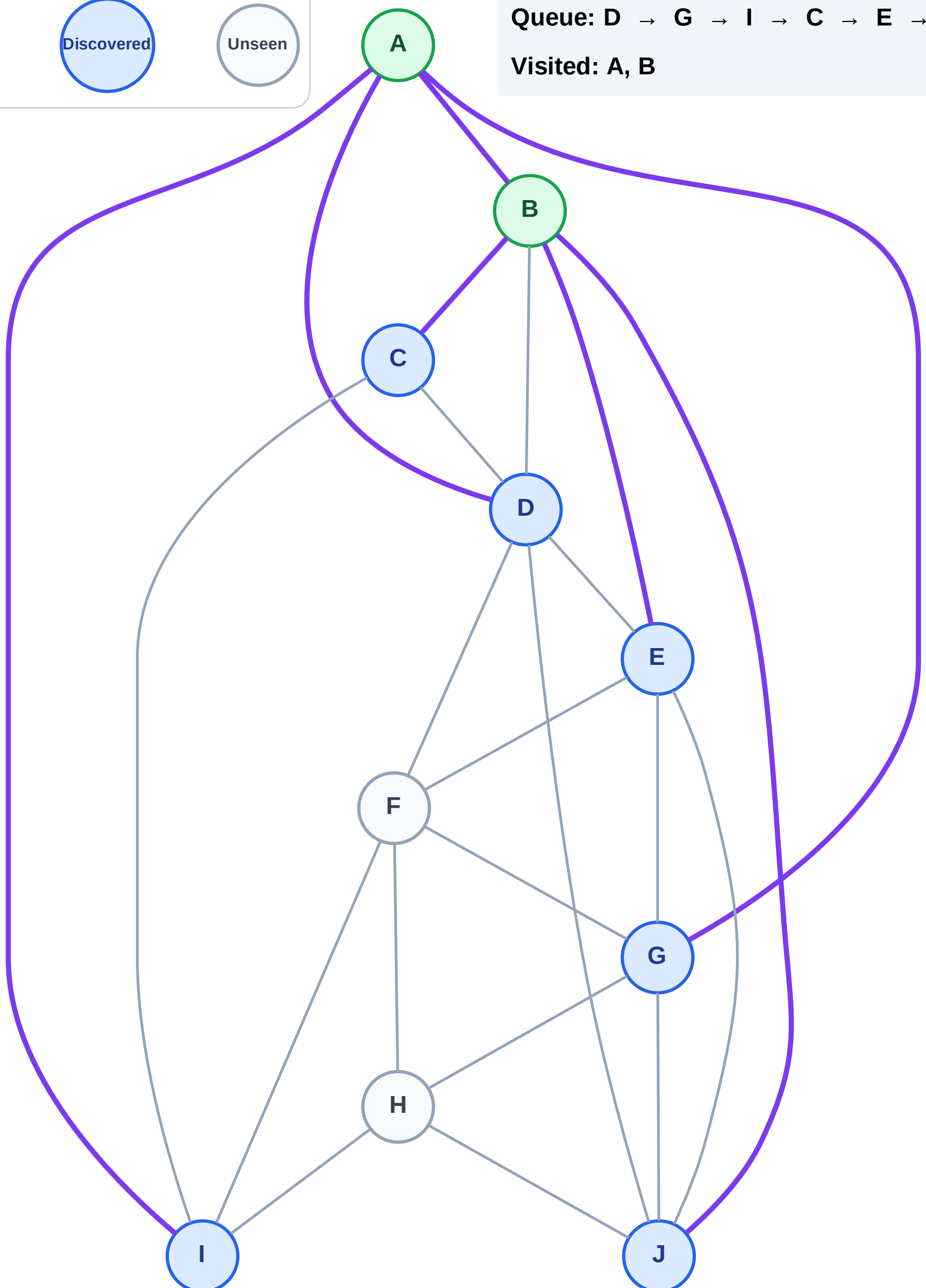
Processing

Discovered

Unseen

Queue: D → G → I → C → E → J

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

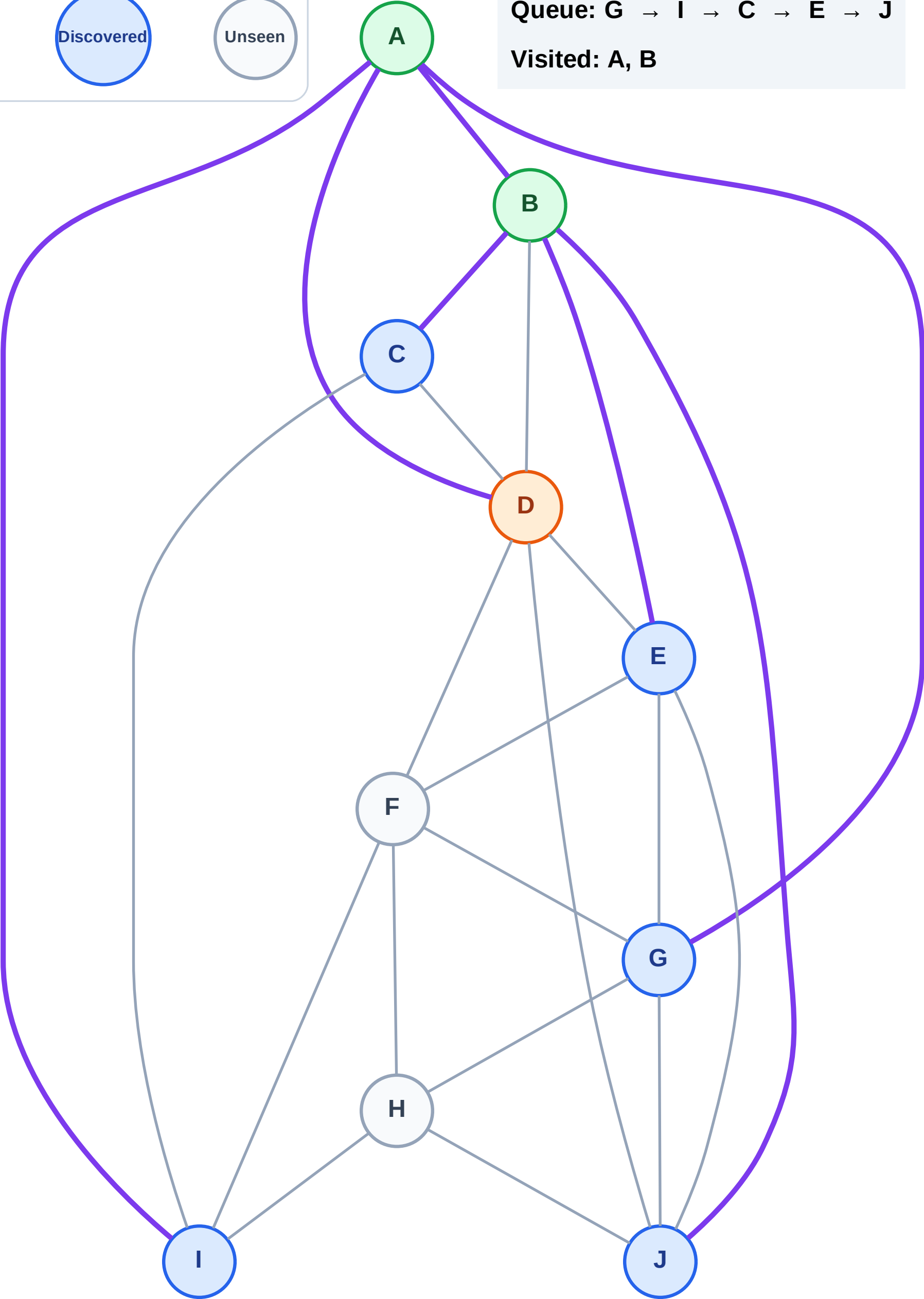
Processing

Discovered

Unseen

Queue: G → I → C → E → J

Visited: A, B



Step 7: Discover F from D

Legend

Unseen

Visited: A, B, D

BFS Traversal

Step 8: Process node G

Legend

Complete

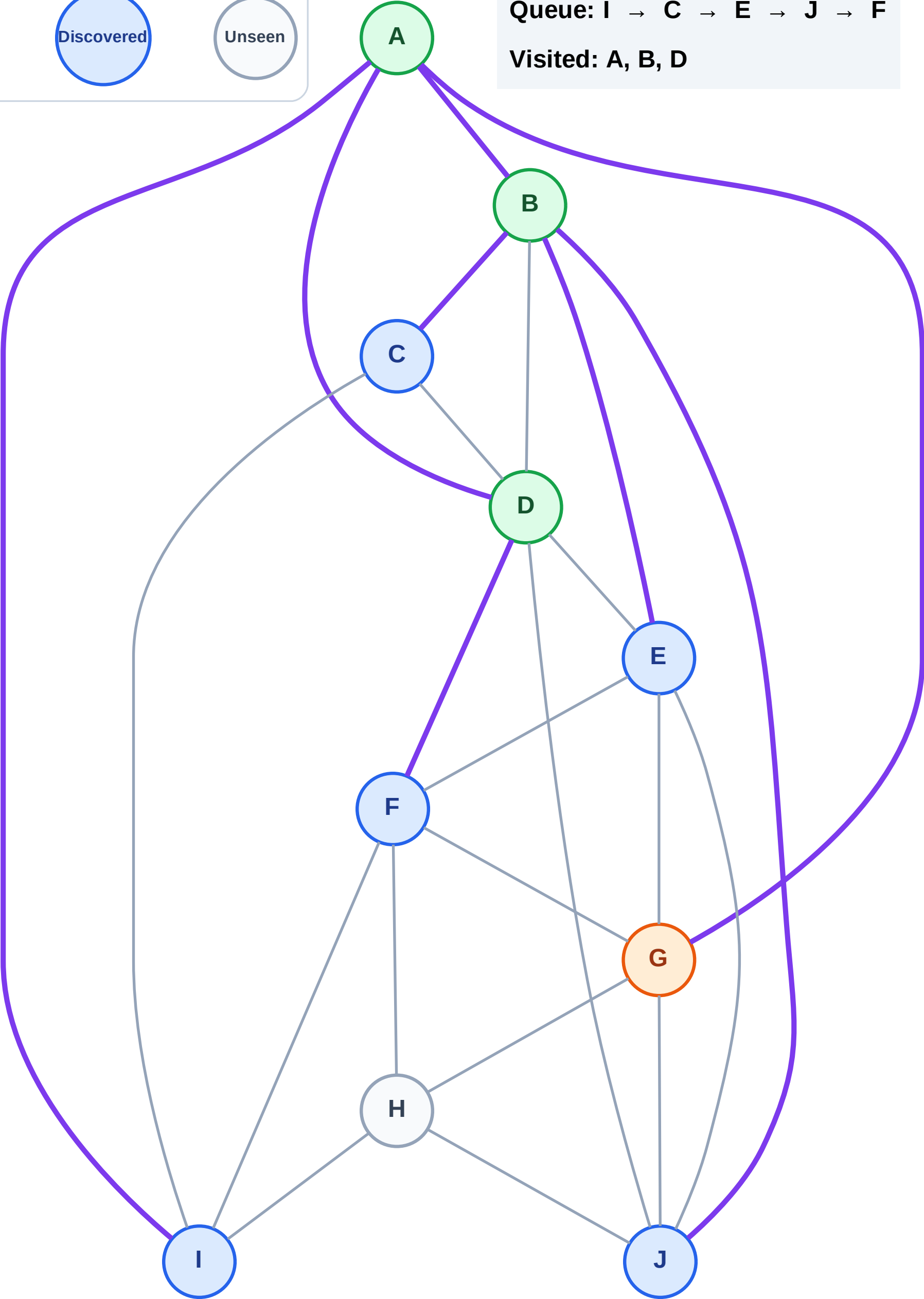
Processing

Discovered

Unseen

Queue: I → C → E → J → F

Visited: A, B, D



BFS Traversal

Step 9: Discover H from G

Legend

Complete

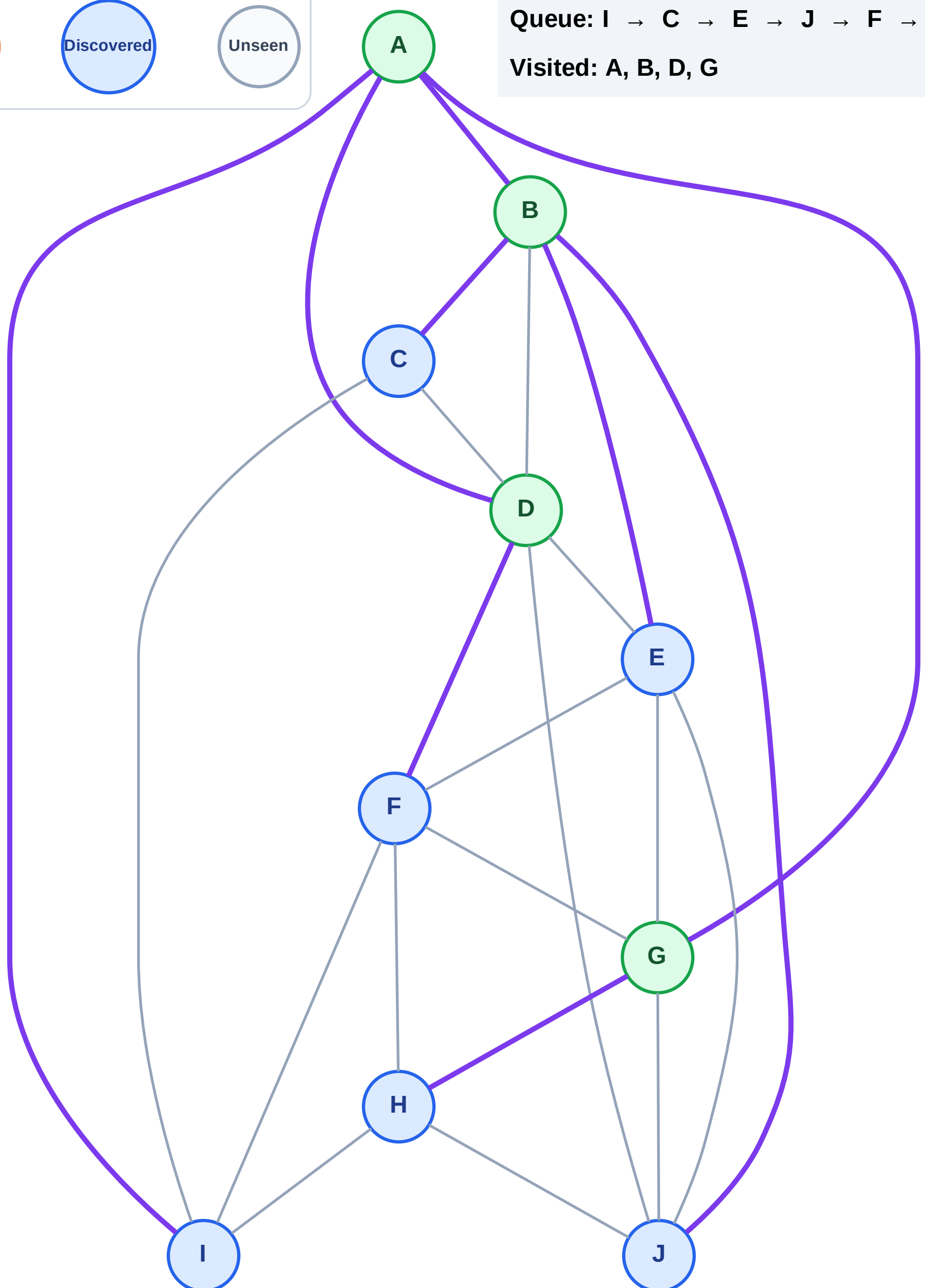
Processing

Discovered

Unseen

Queue: I → C → E → J → F → H

Visited: A, B, D, G



BFS Traversal

Step 10: Process node I

Legend

Complete

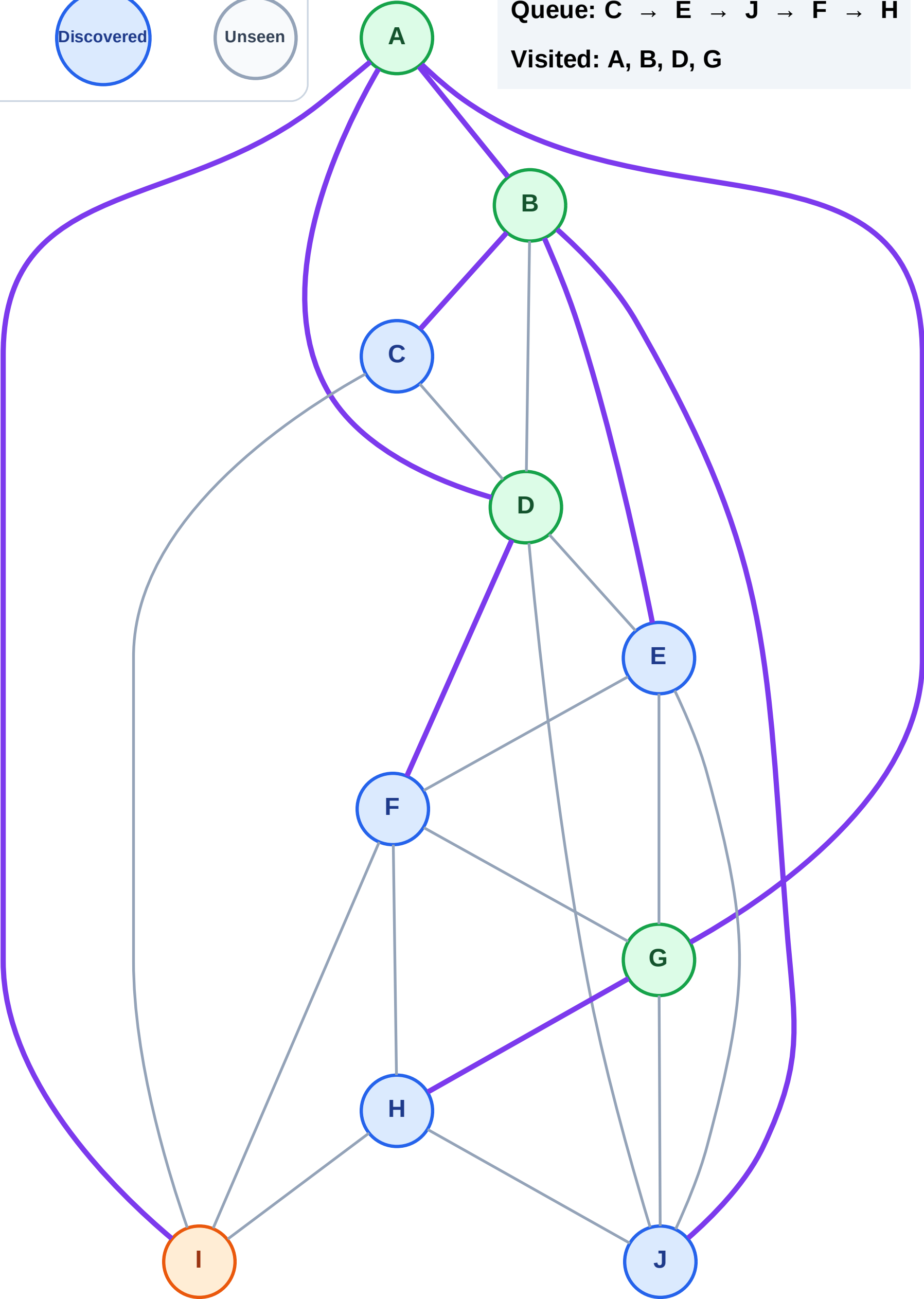
Processing

Discovered

Unseen

Queue: C → E → J → F → H

Visited: A, B, D, G



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

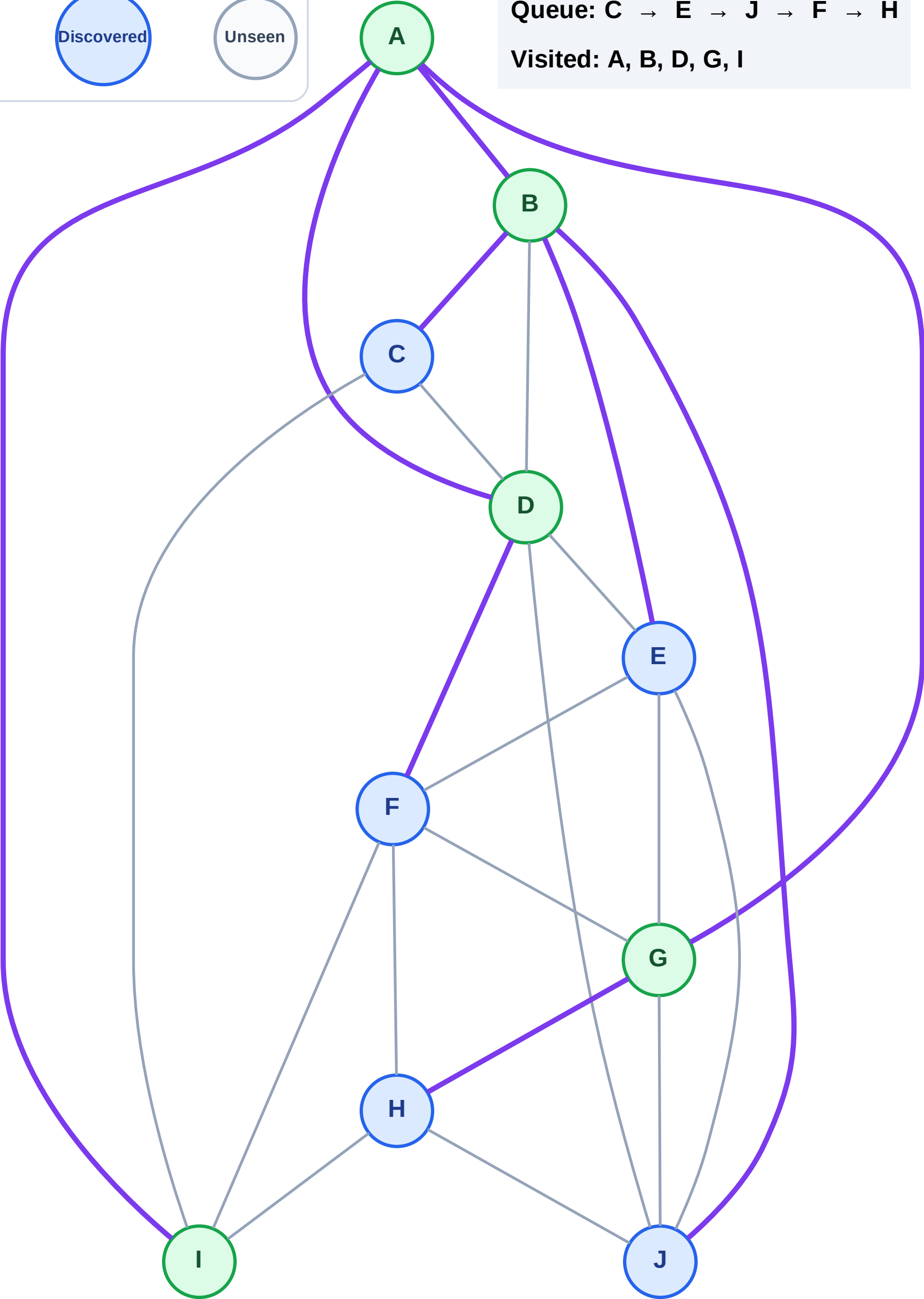
Processing

Discovered

Unseen

Queue: C → E → J → F → H

Visited: A, B, D, G, I



BFS Traversal

Step 12: Process node C

Legend

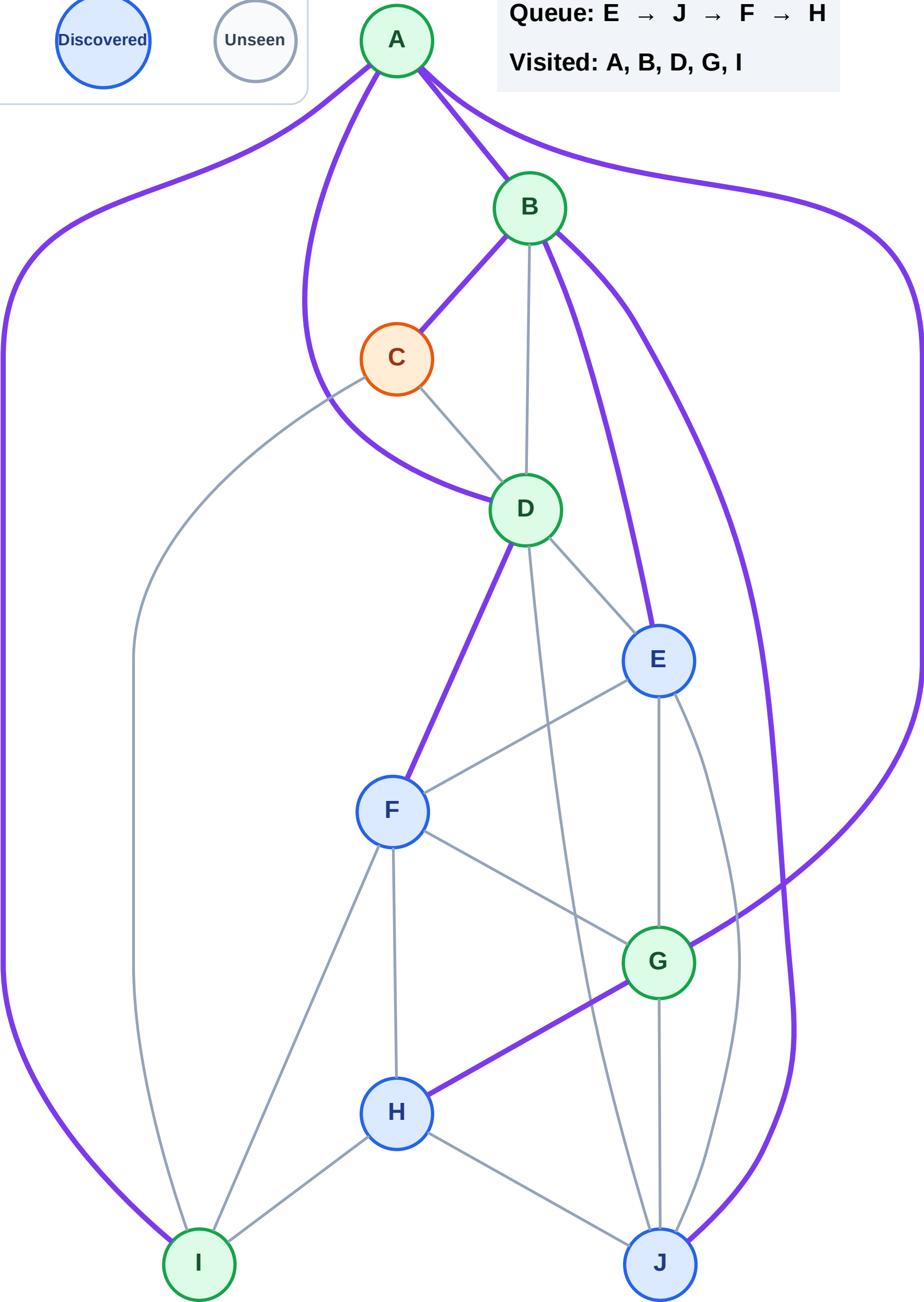
Complete

Processing

Discovered

Unseen

Queue: E → J → F → H
Visited: A, B, D, G, I



BFS Traversal

Step 13: No new neighbors from C

Legend

Complete

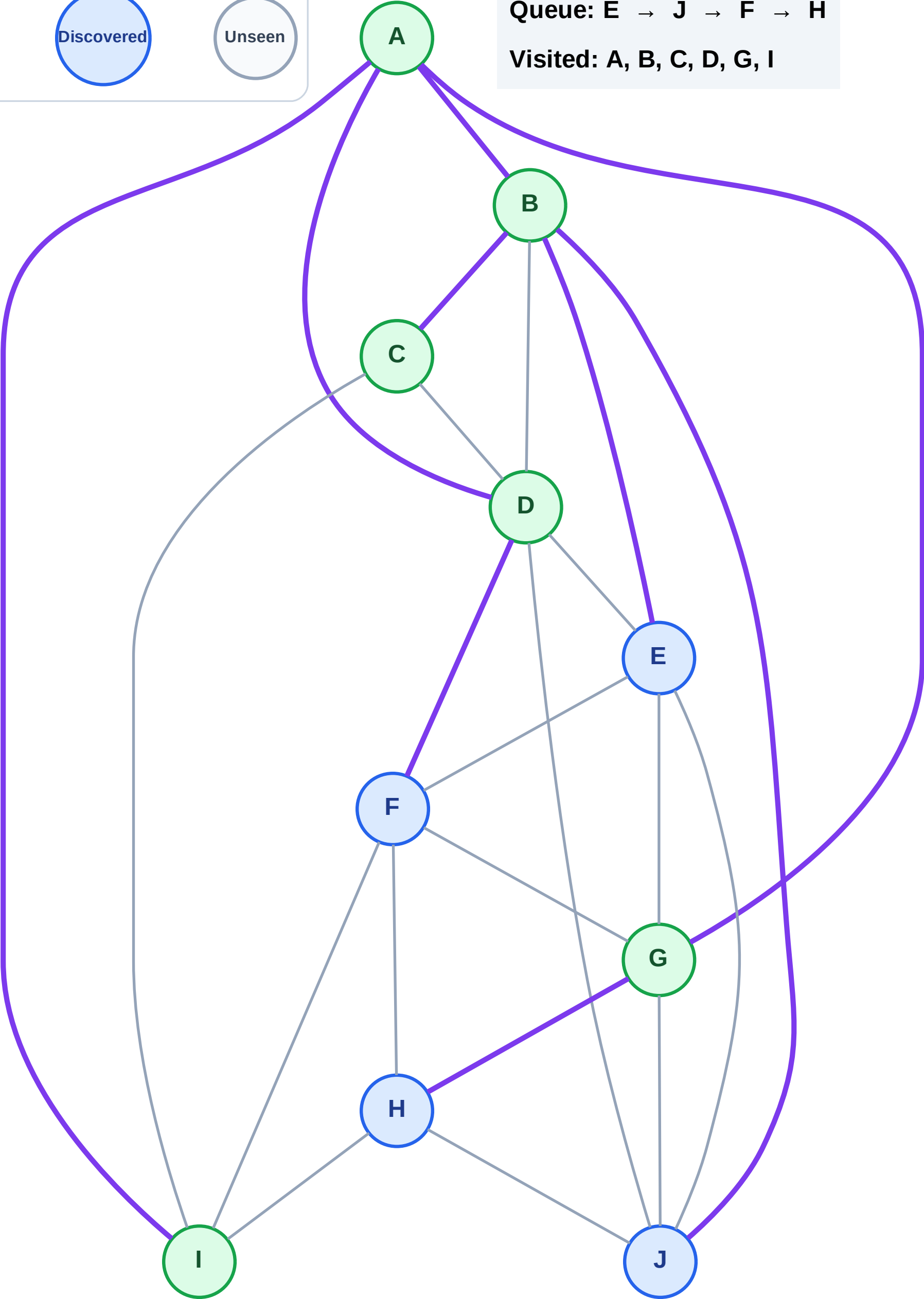
Processing

Discovered

Unseen

Queue: E → J → F → H

Visited: A, B, C, D, G, I



BFS Traversal

Step 14: Process node E

Legend

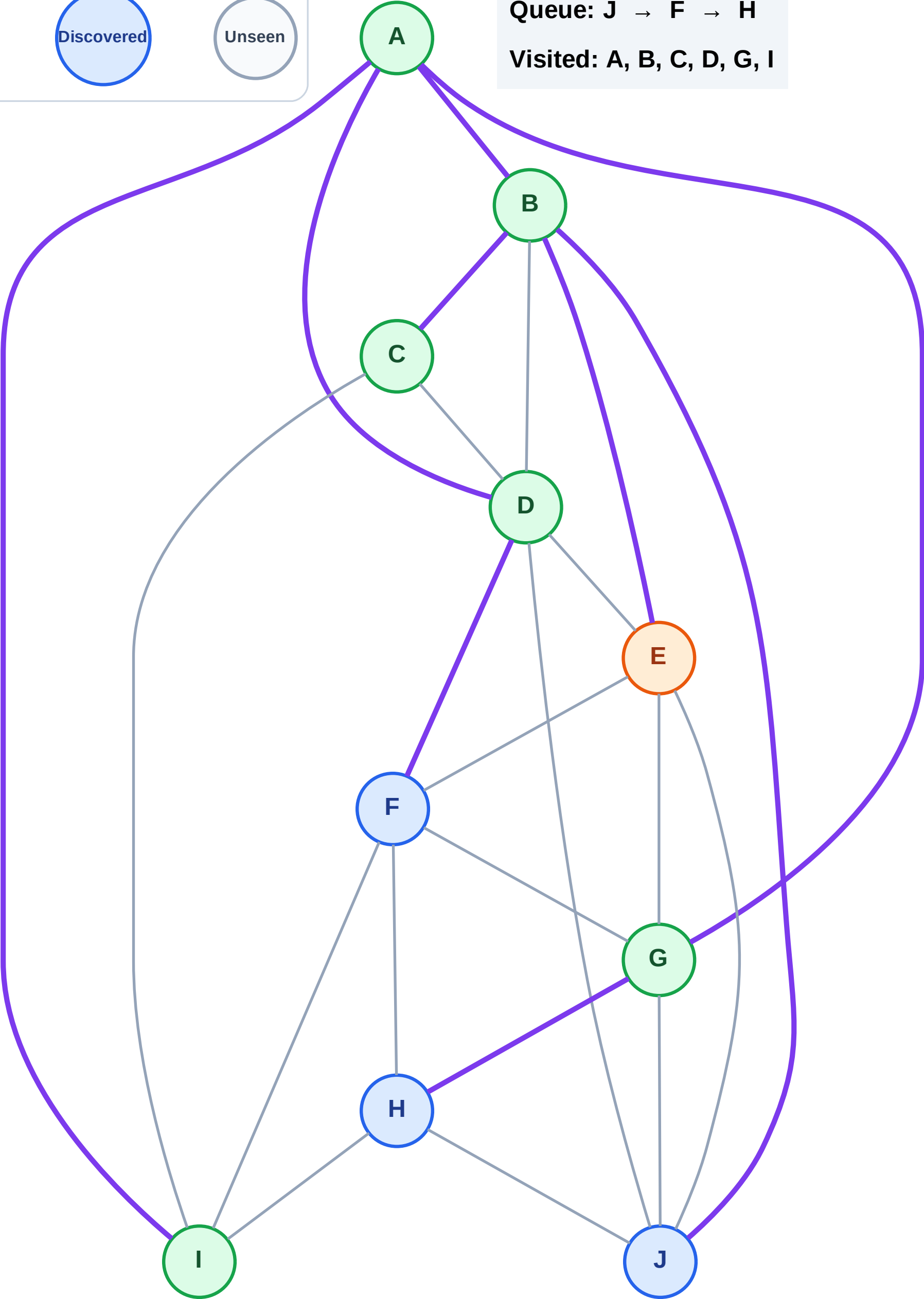
Complete

Processing

Discovered

Unseen

Queue: J → F → H
Visited: A, B, C, D, G, I



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

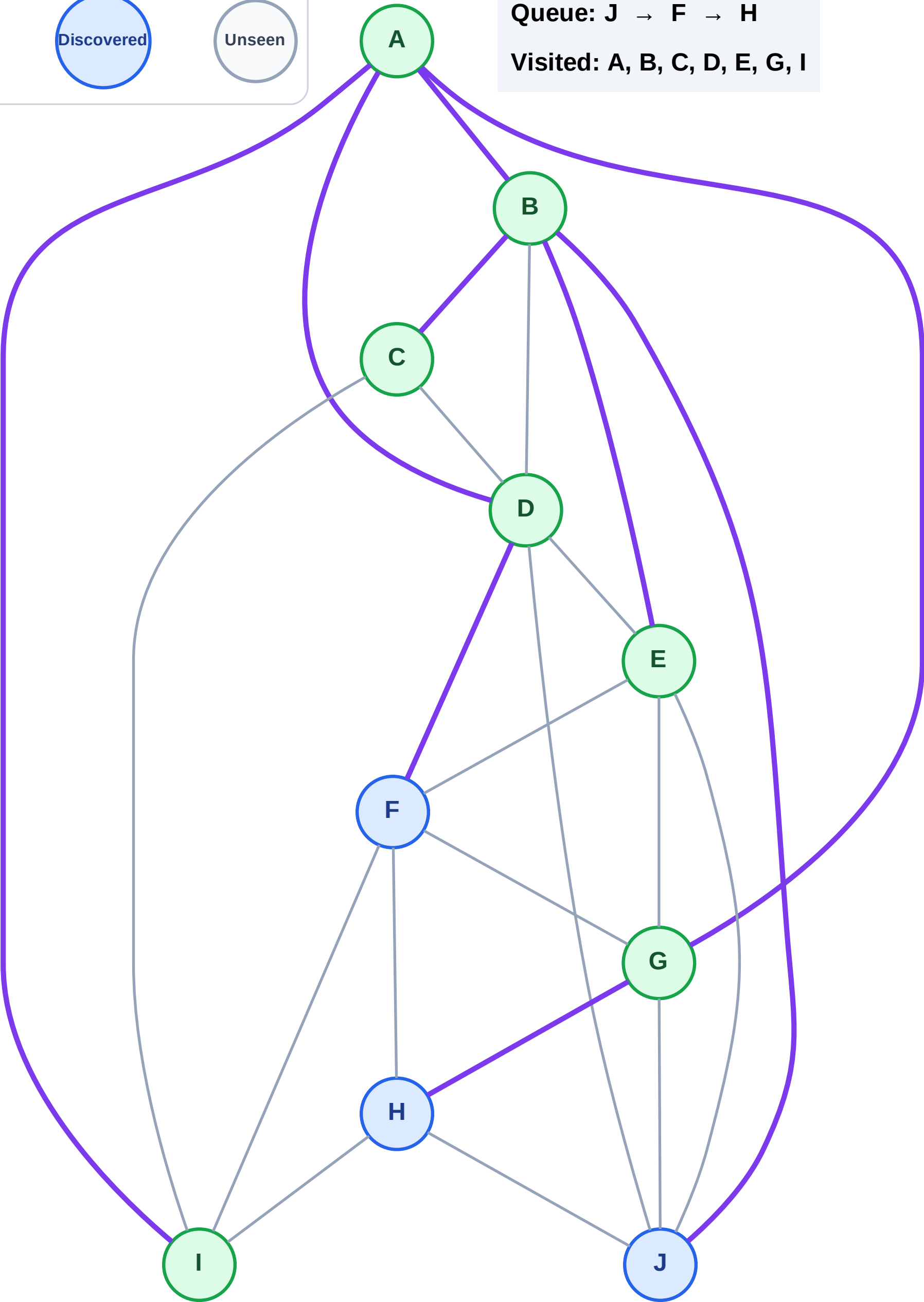
Processing

Discovered

Unseen

Queue: J → F → H

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node J

Legend

Complete

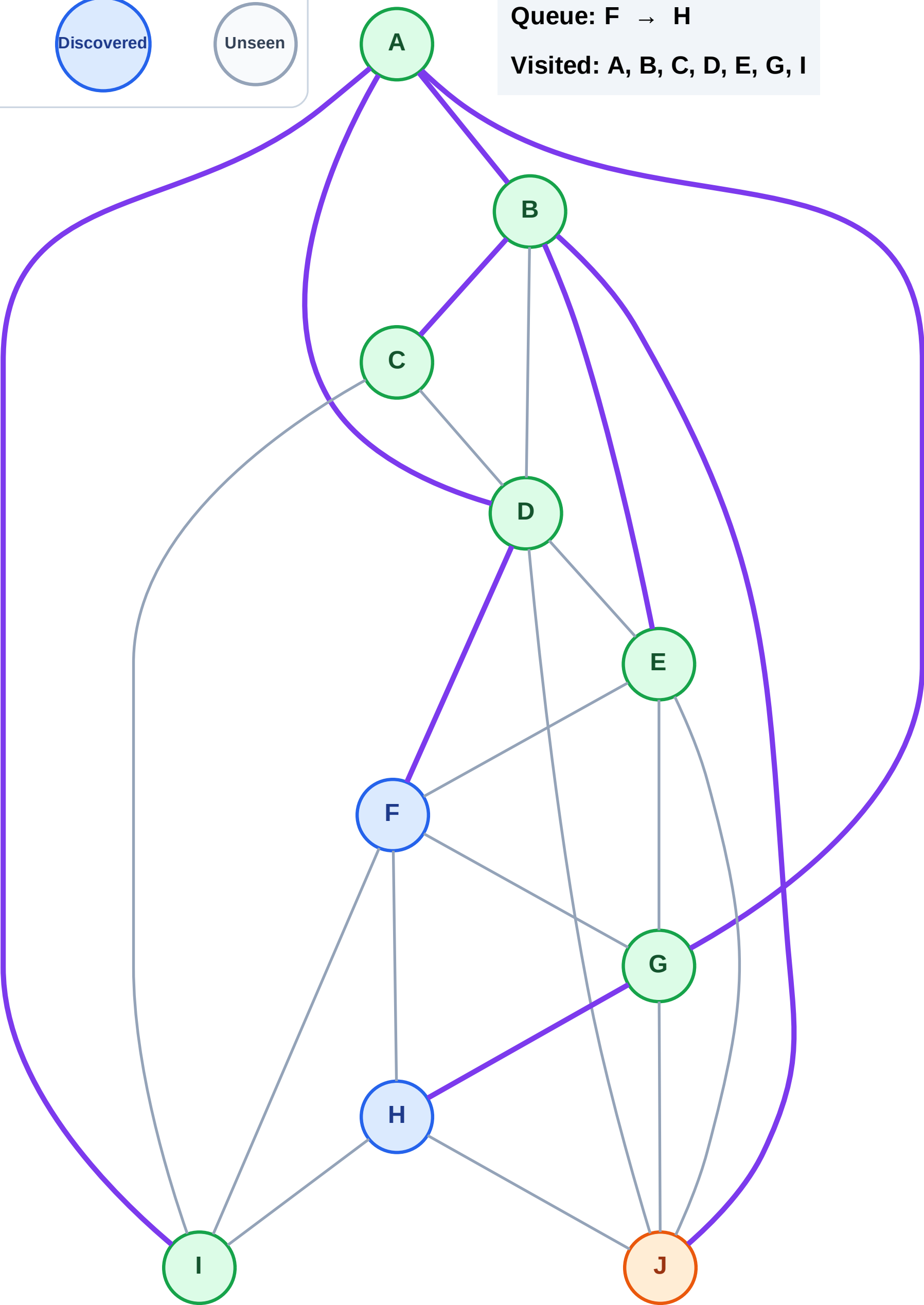
Processing

Discovered

Unseen

Queue: F → H

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 17: No new neighbors from J

Legend

Complete

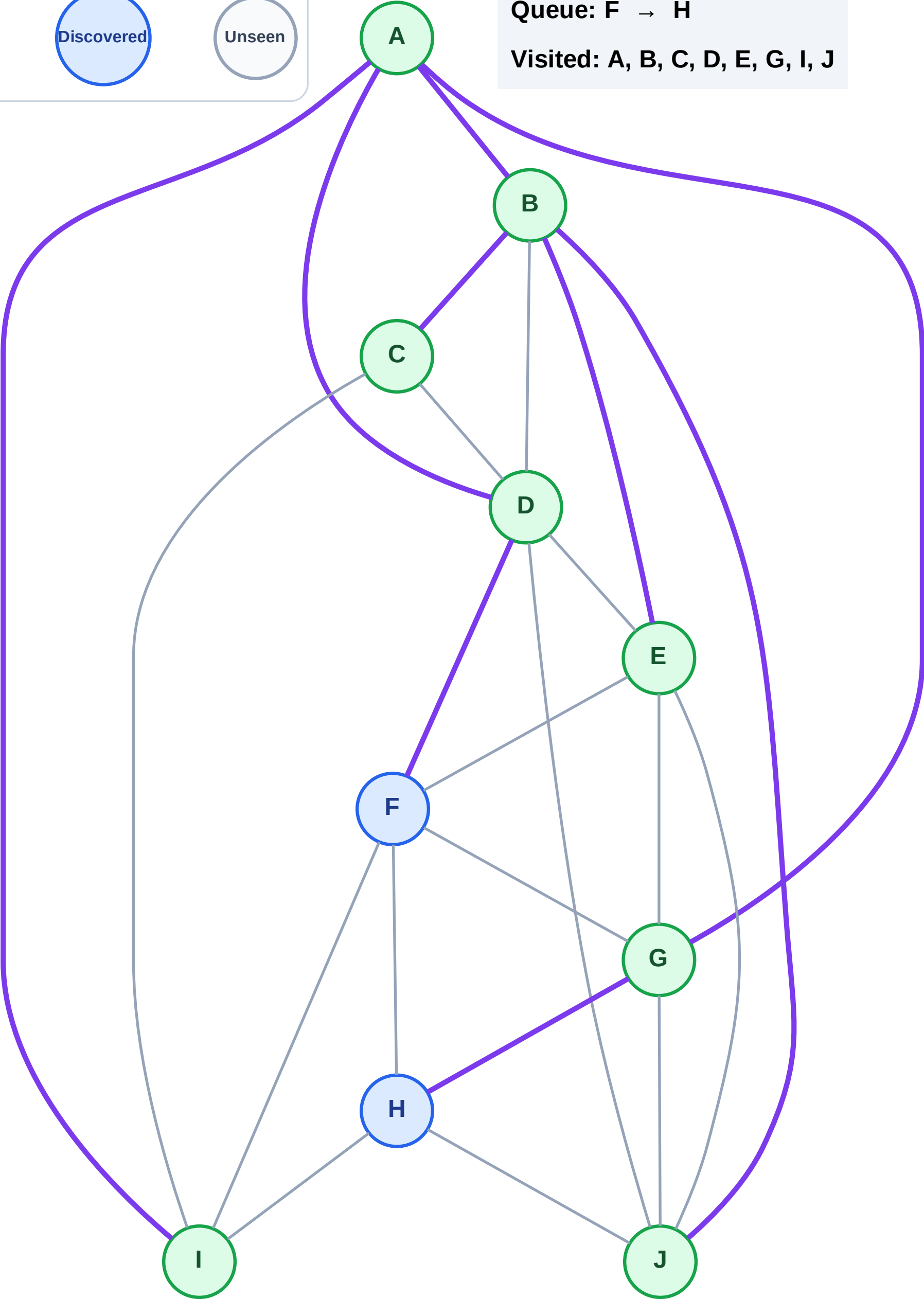
Processing

Discovered

Unseen

Queue: F → H

Visited: A, B, C, D, E, G, I, J



BFS Traversal

Step 18: Process node F

Legend

Complete

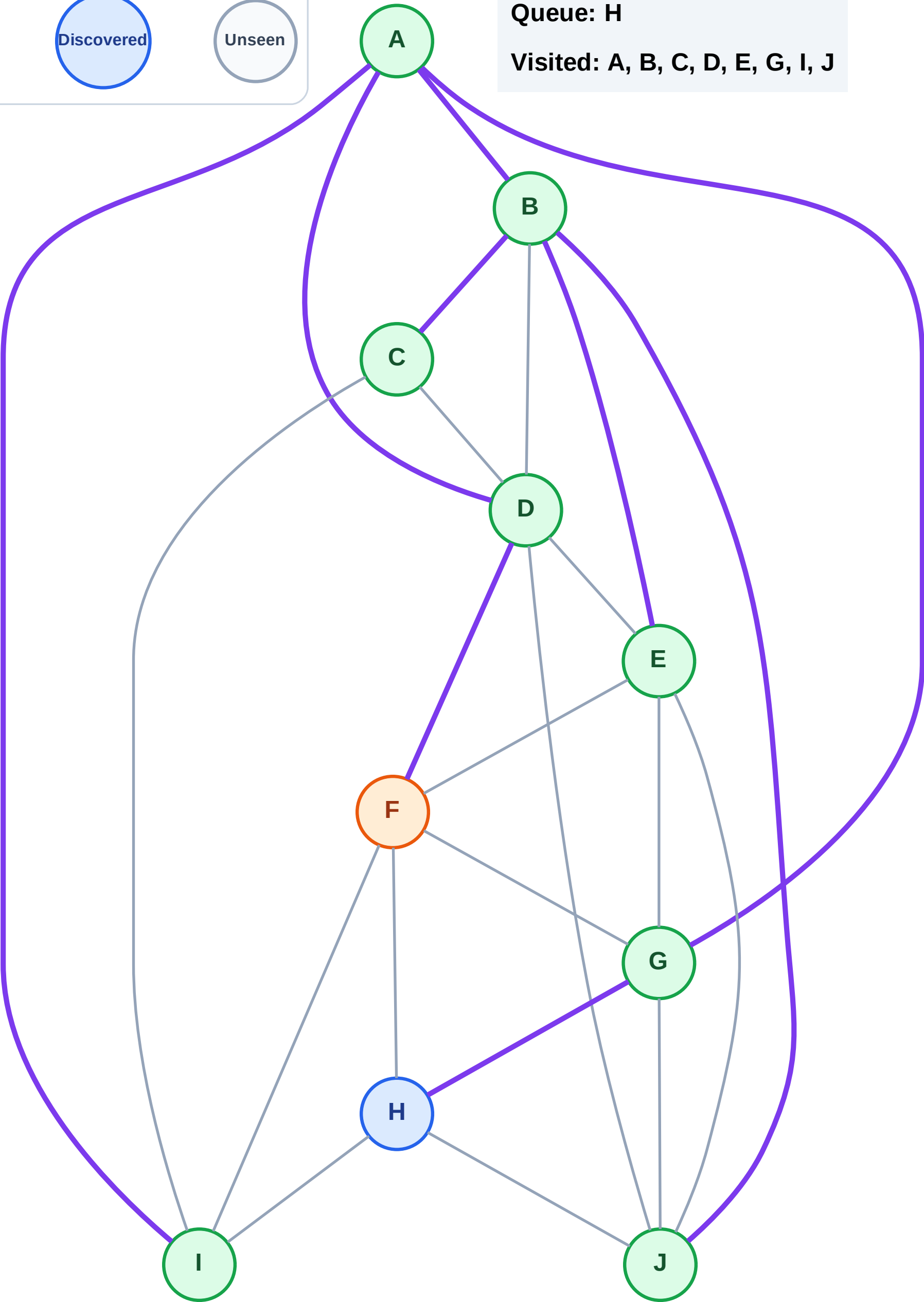
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, G, I, J



BFS Traversal

Step 19: No new neighbors from F

Legend

Complete

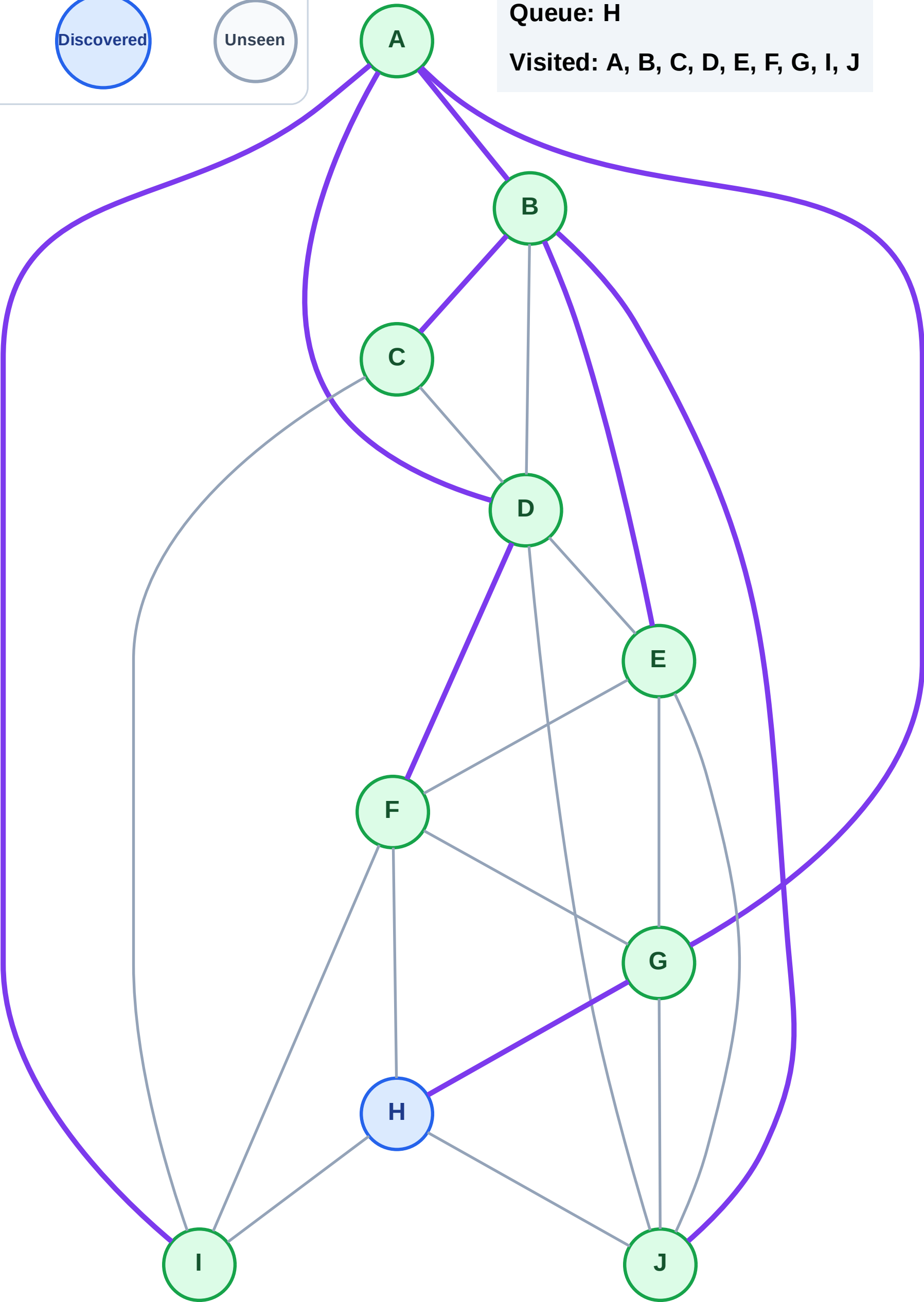
Processing

Discovered

Unseen

Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

Complete

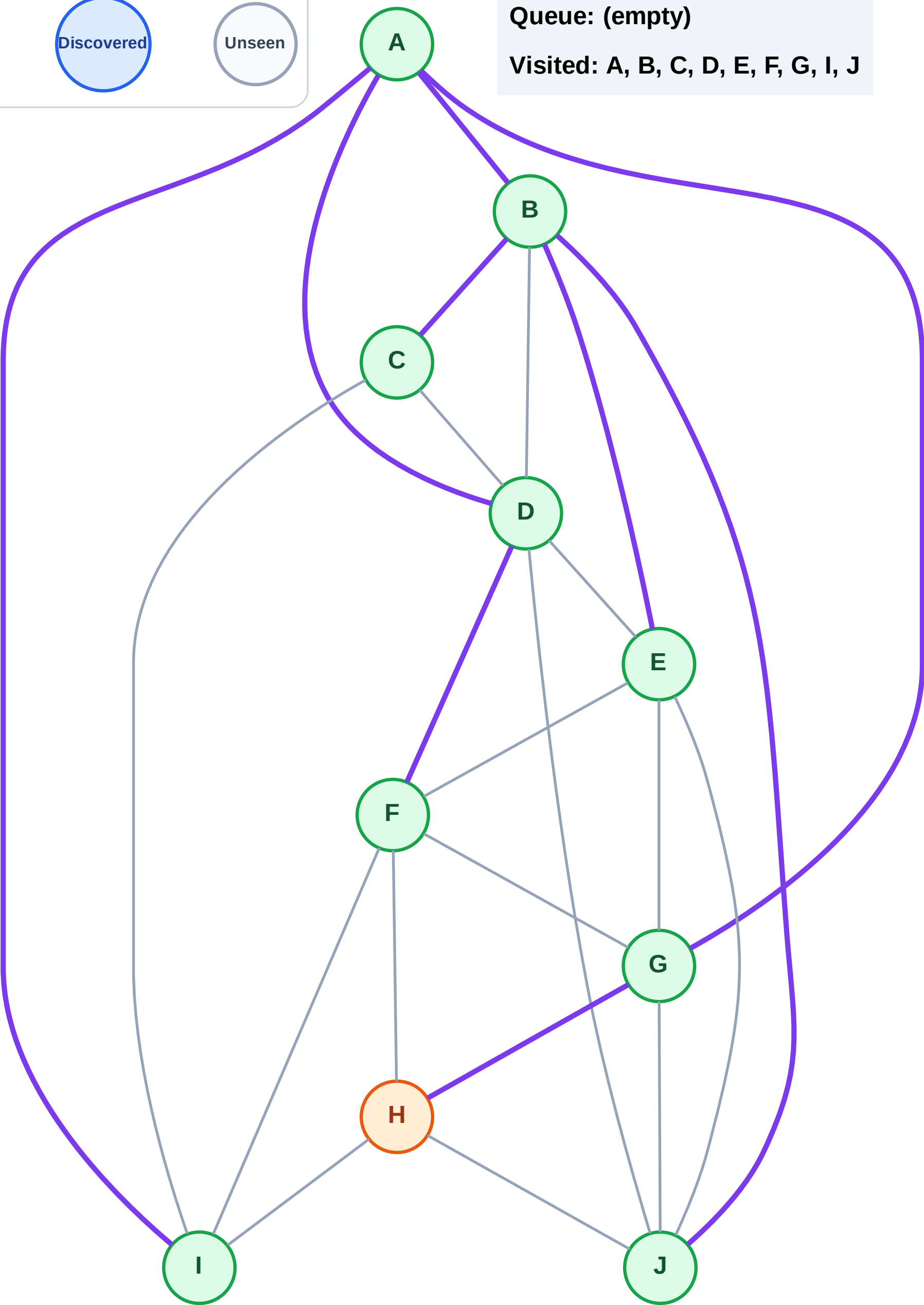
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

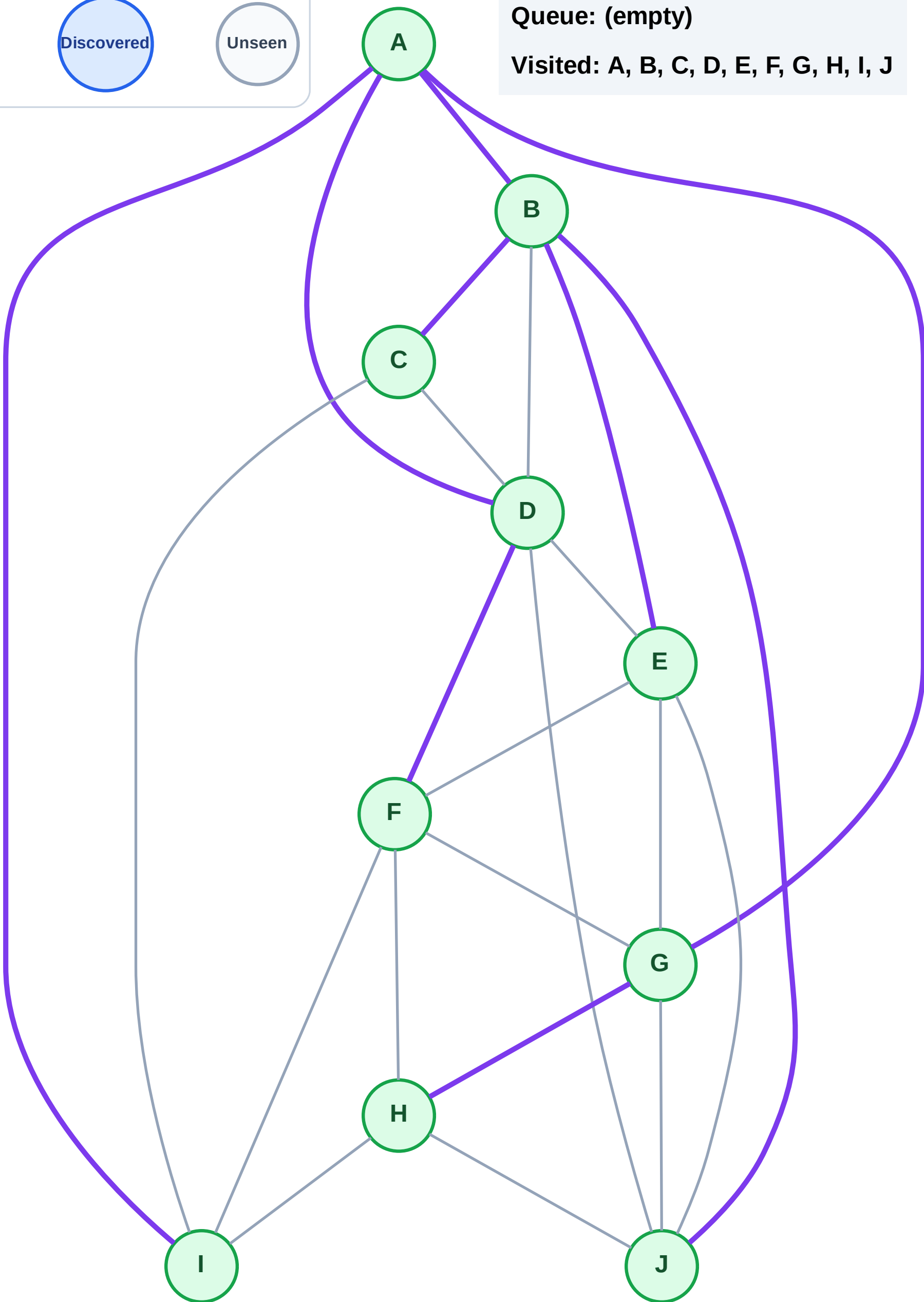
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

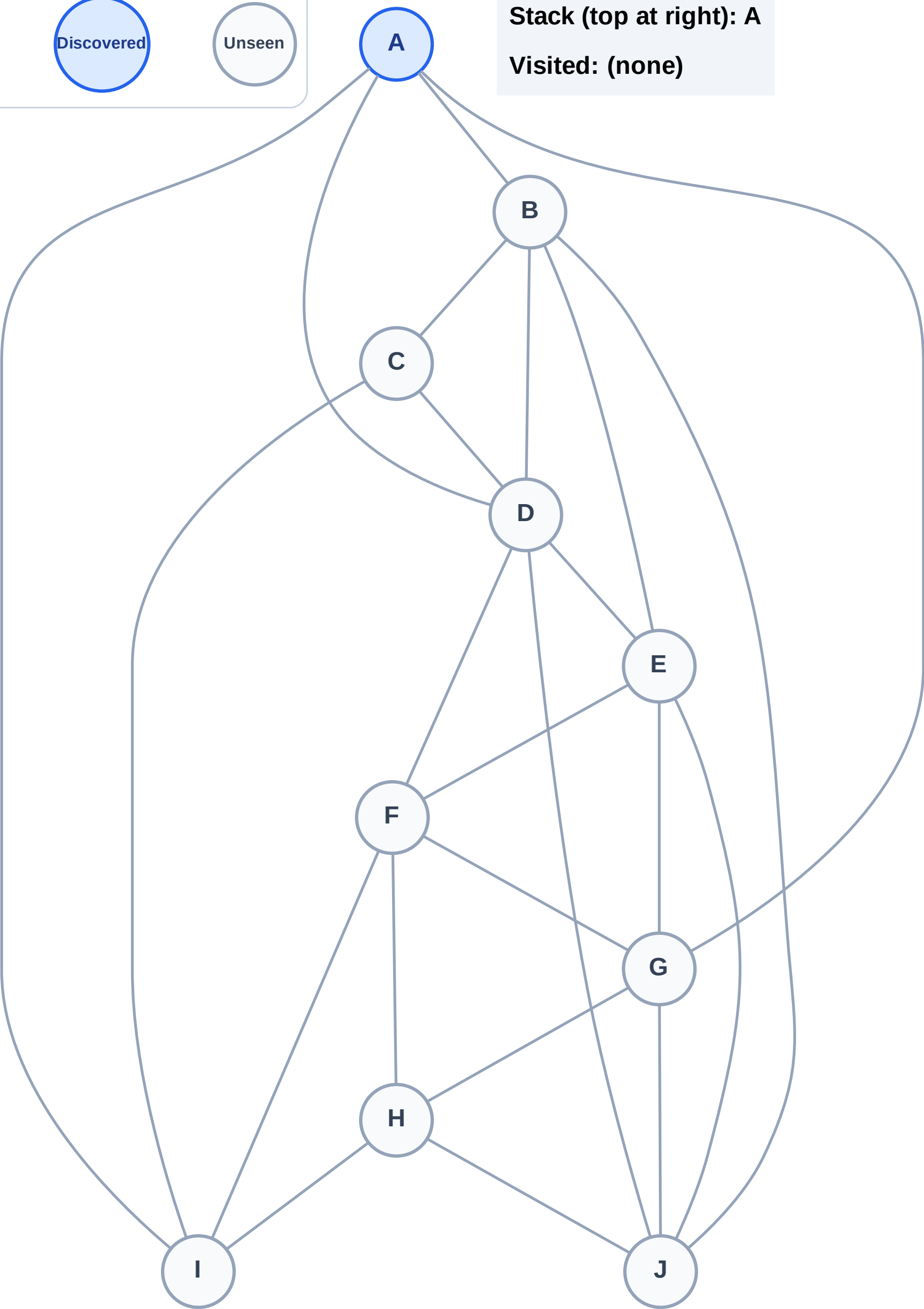
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

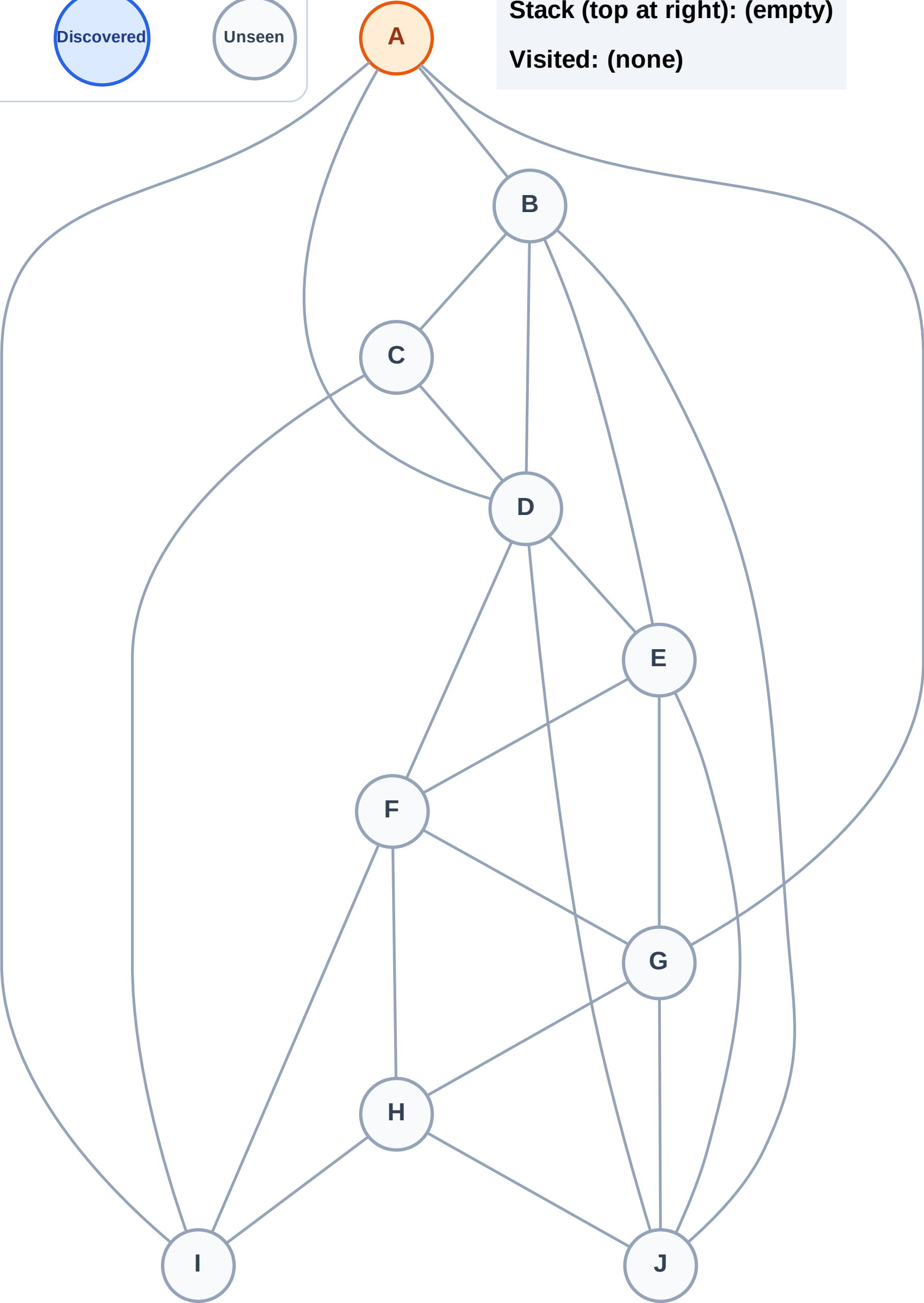
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover I, G, D, B from A

Legend

Complete

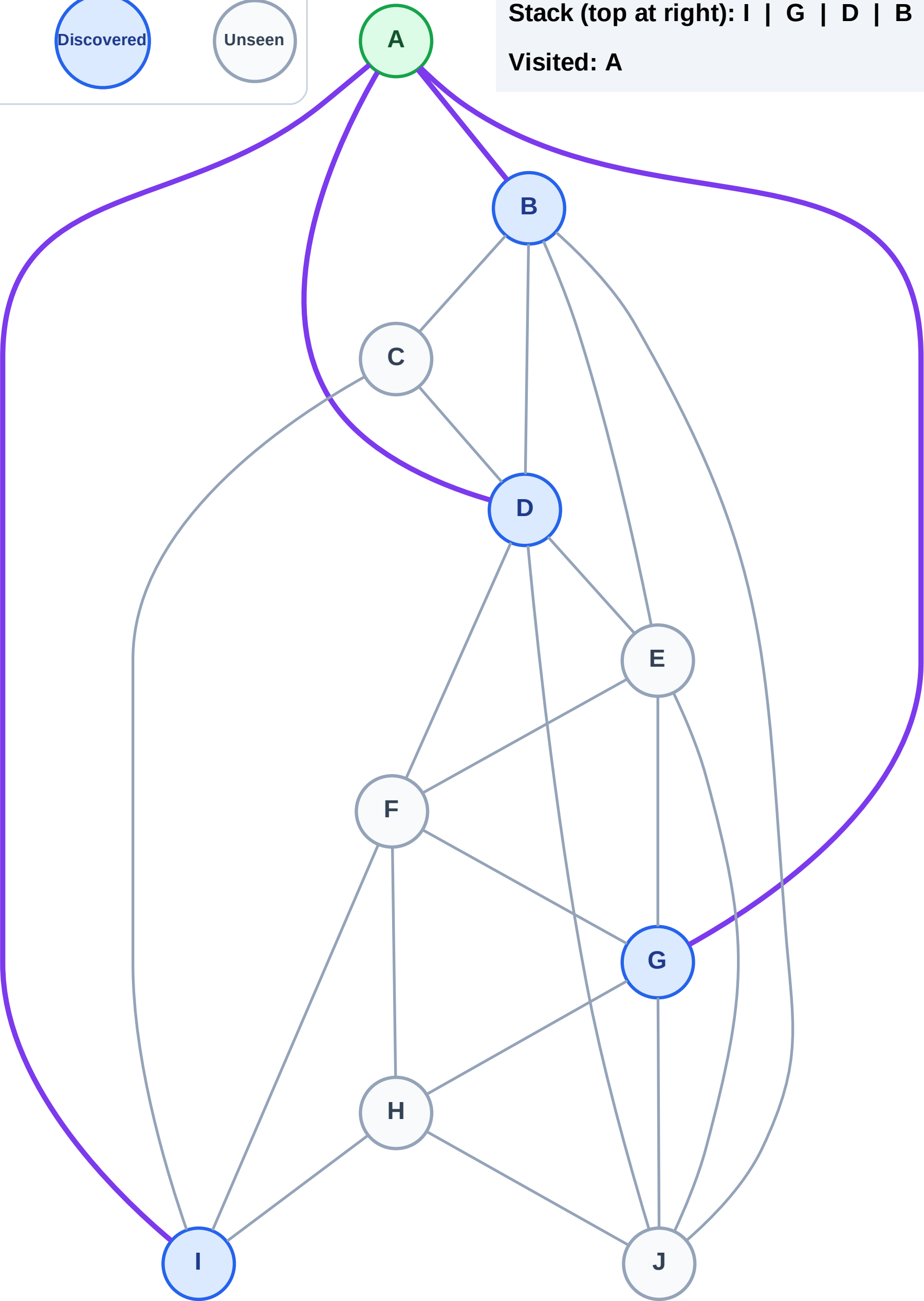
Processing

Discovered

Unseen

Stack (top at right): I | G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

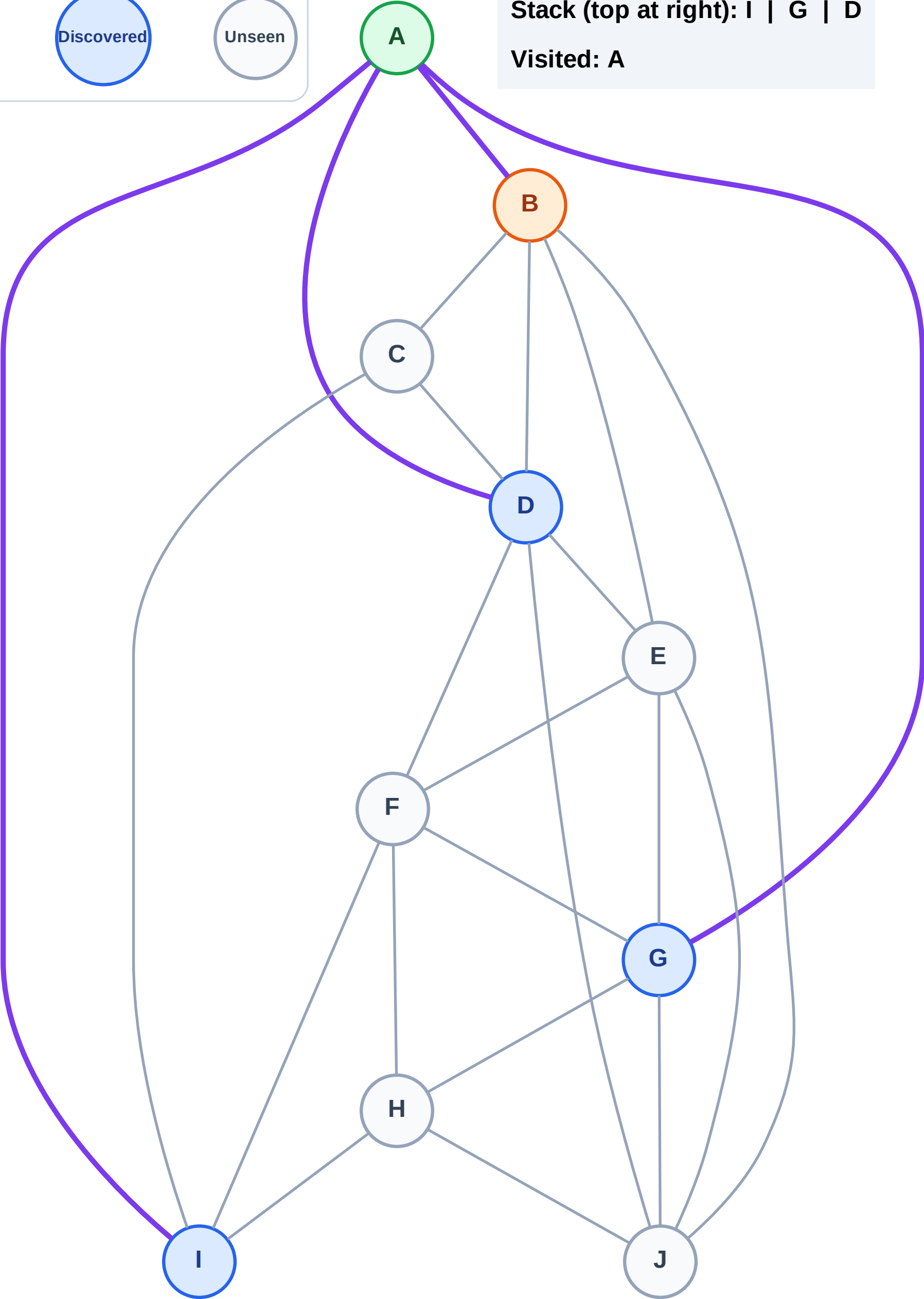
Processing

Discovered

Unseen

Stack (top at right): I | G | D

Visited: A



DFS Traversal

Step 5: Discover J, E, C from B

Legend

Complete

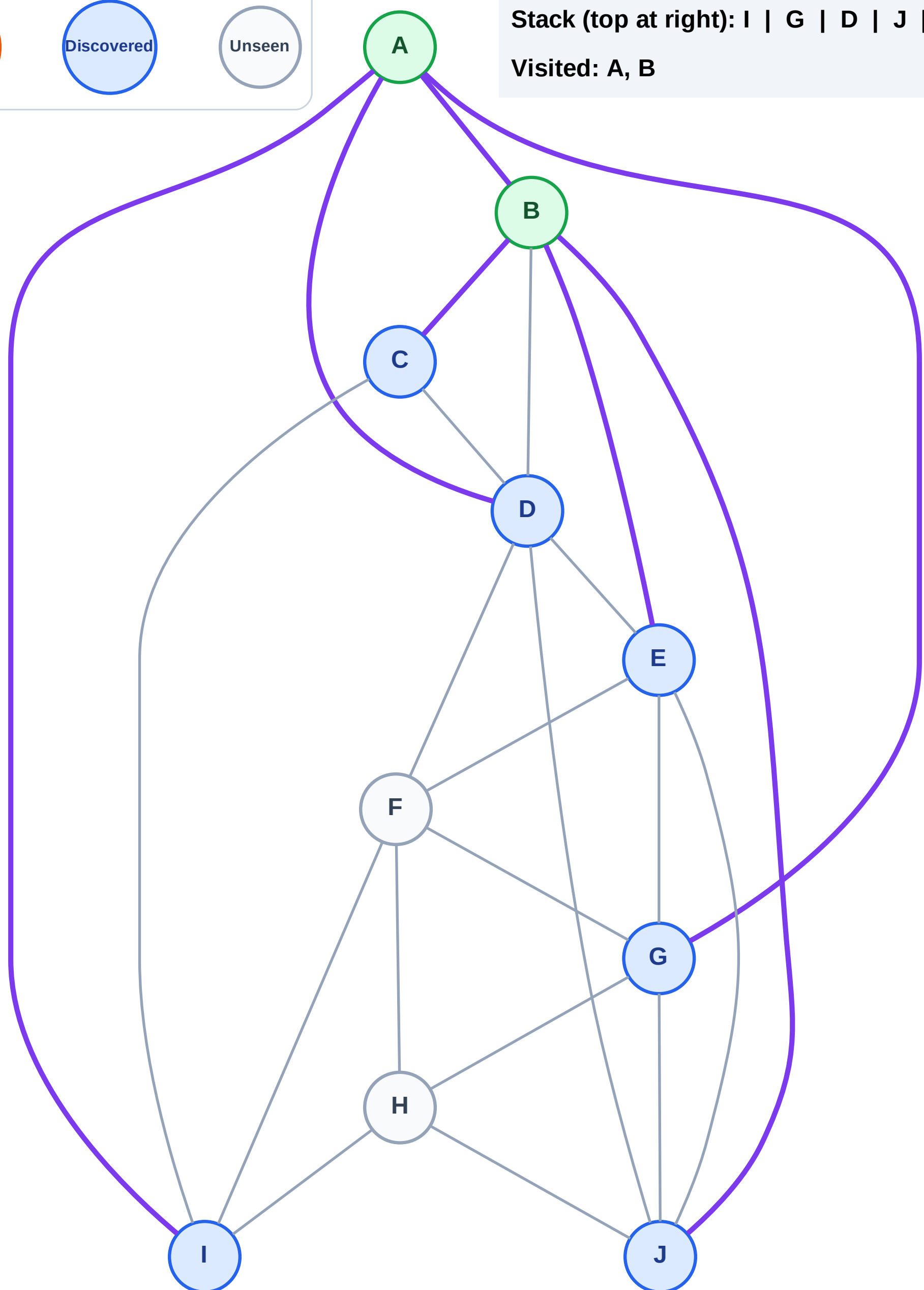
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

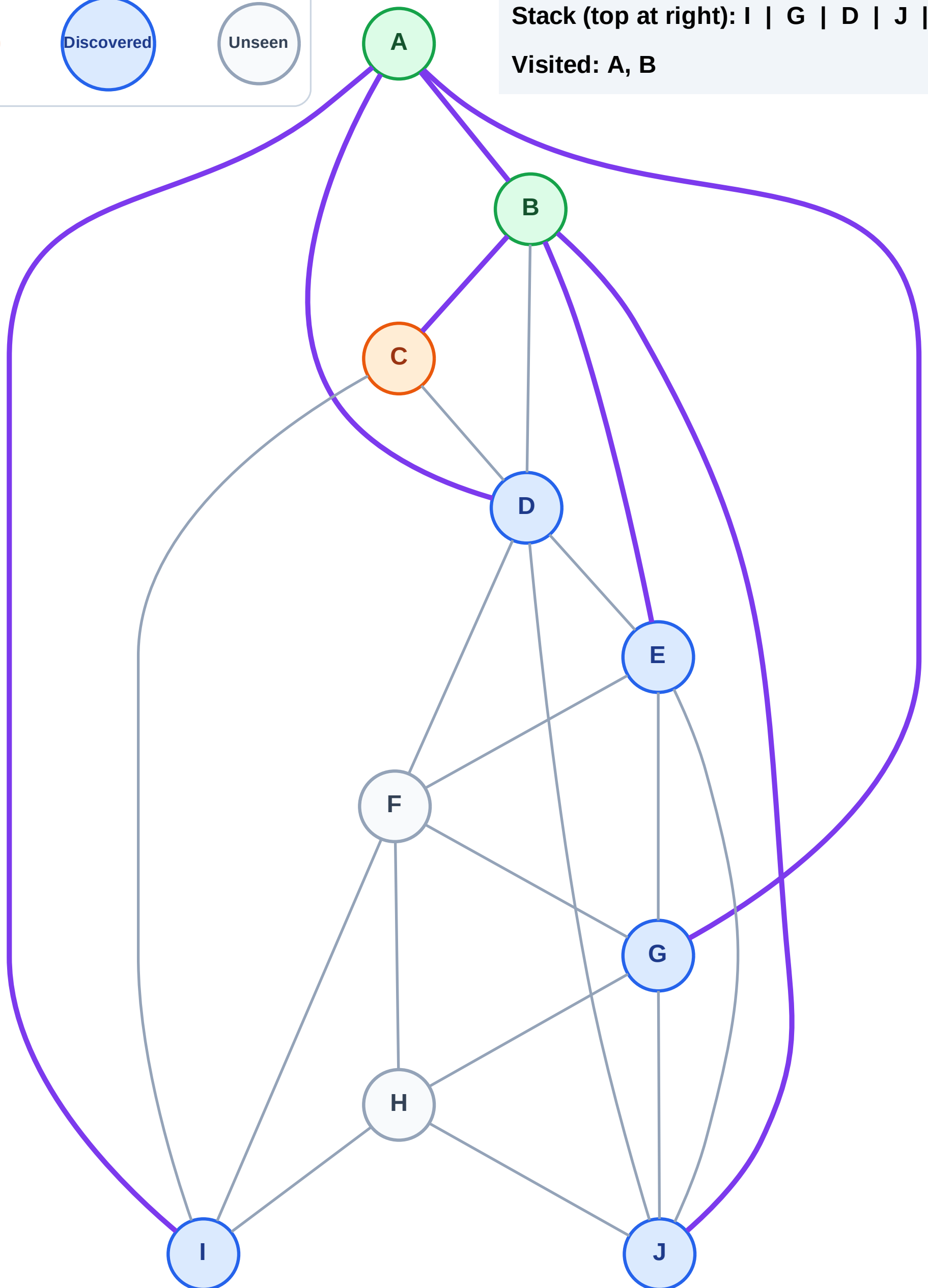
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | E

Visited: A, B



DFS Traversal

Step 7: No new neighbors from C

Legend

Complete

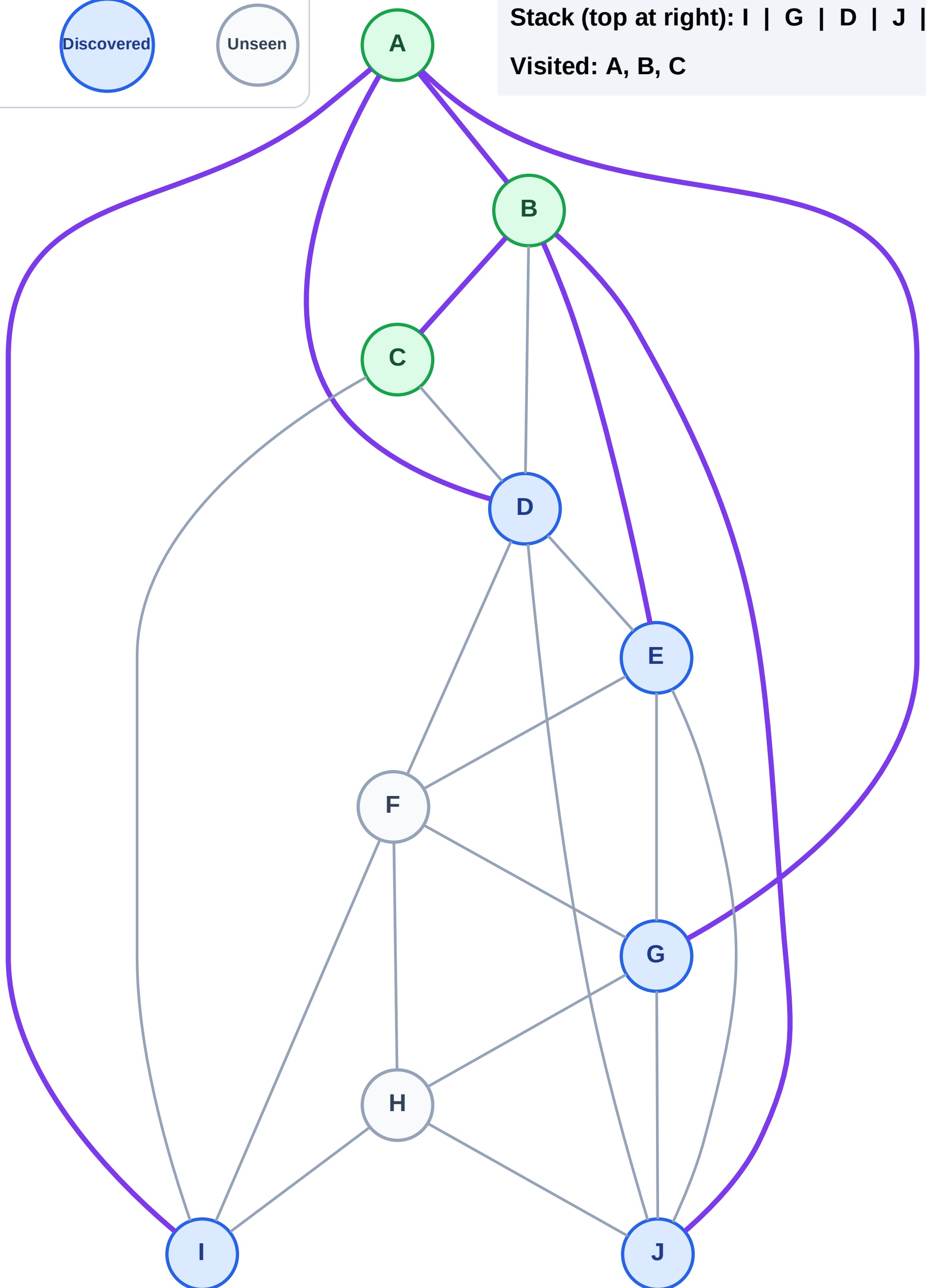
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

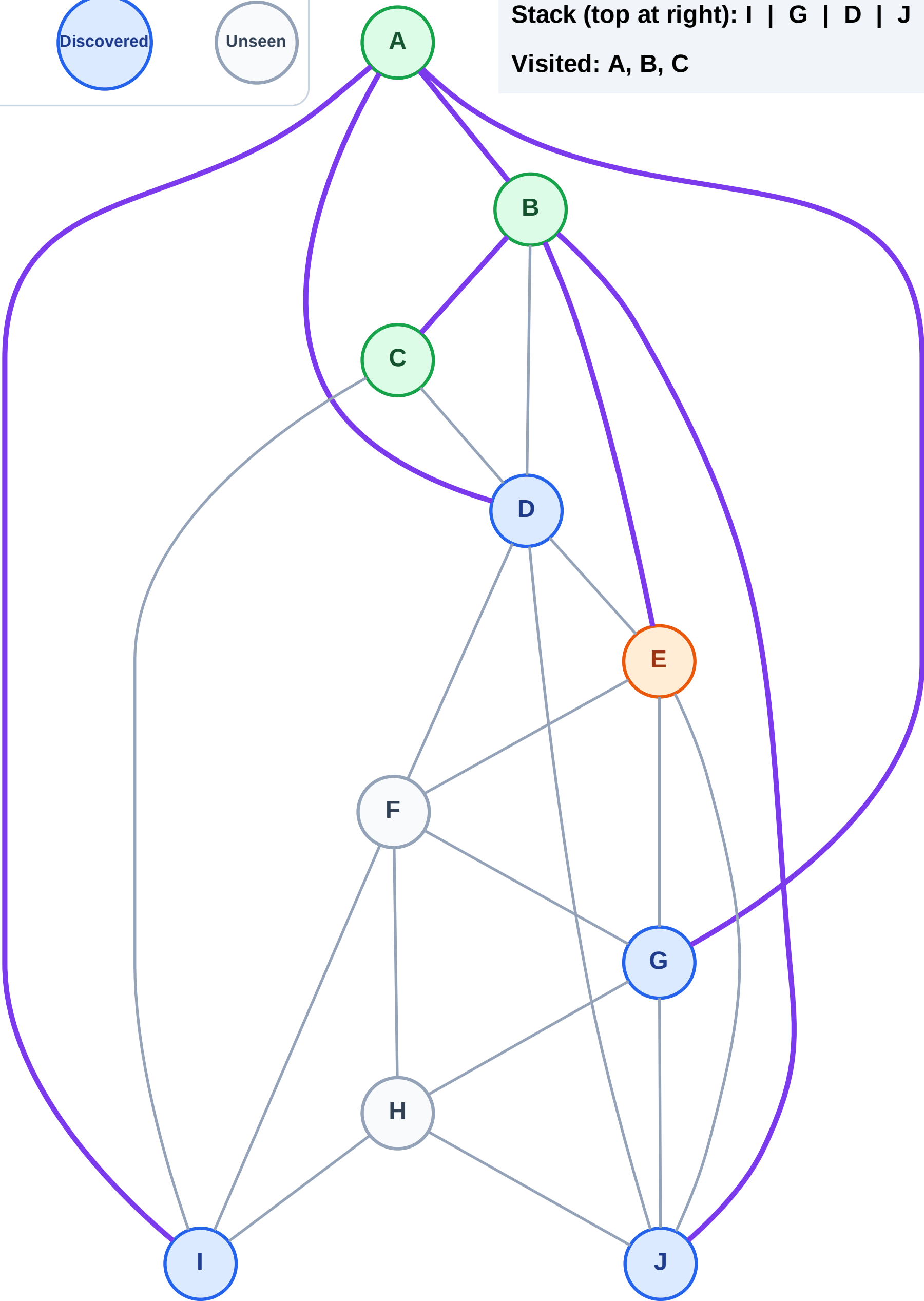
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J

Visited: A, B, C



DFS Traversal

Step 9: Discover F from E

Legend

Complete

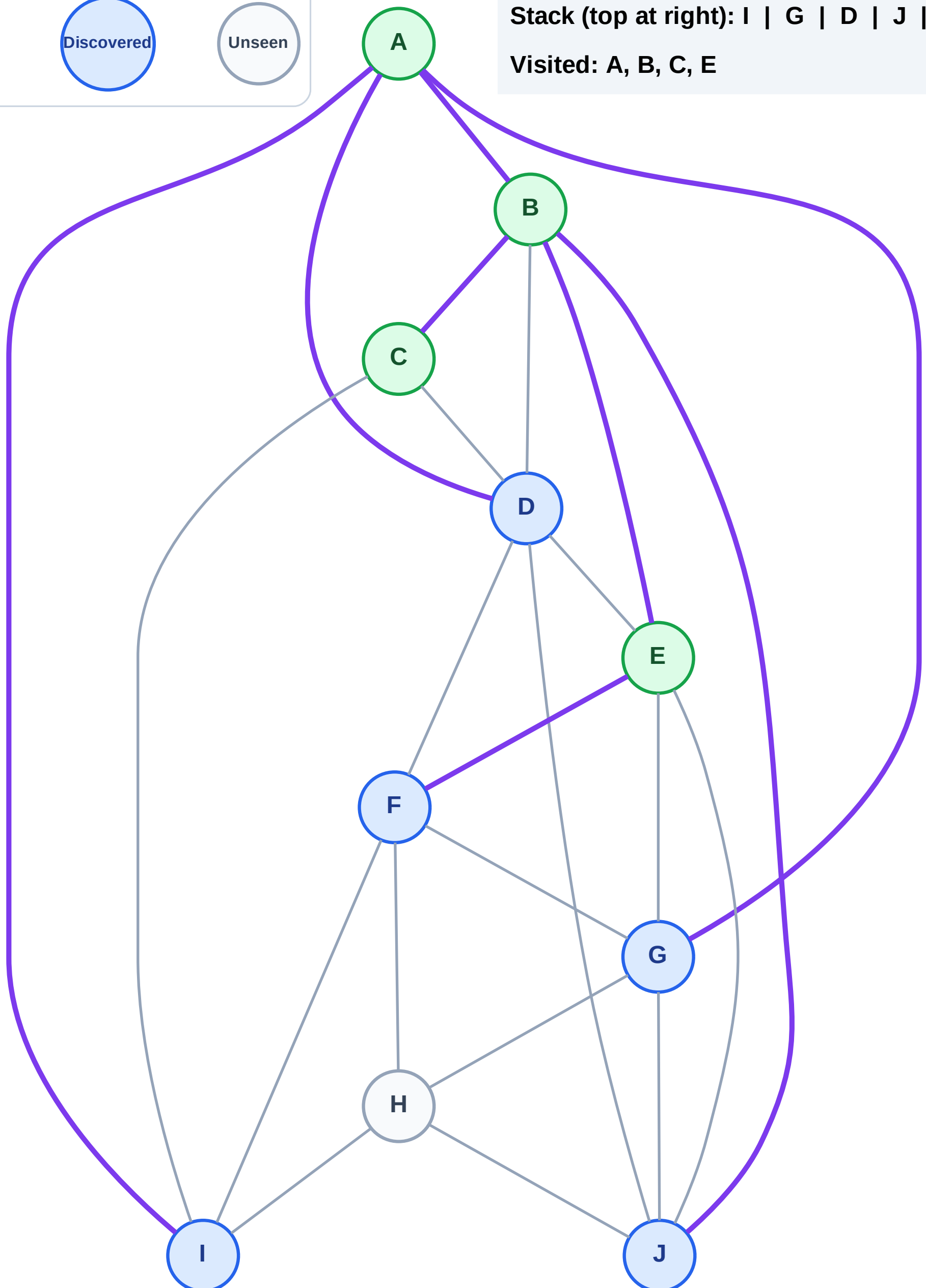
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | F

Visited: A, B, C, E



DFS Traversal

Step 10: Process node F

Legend

Complete

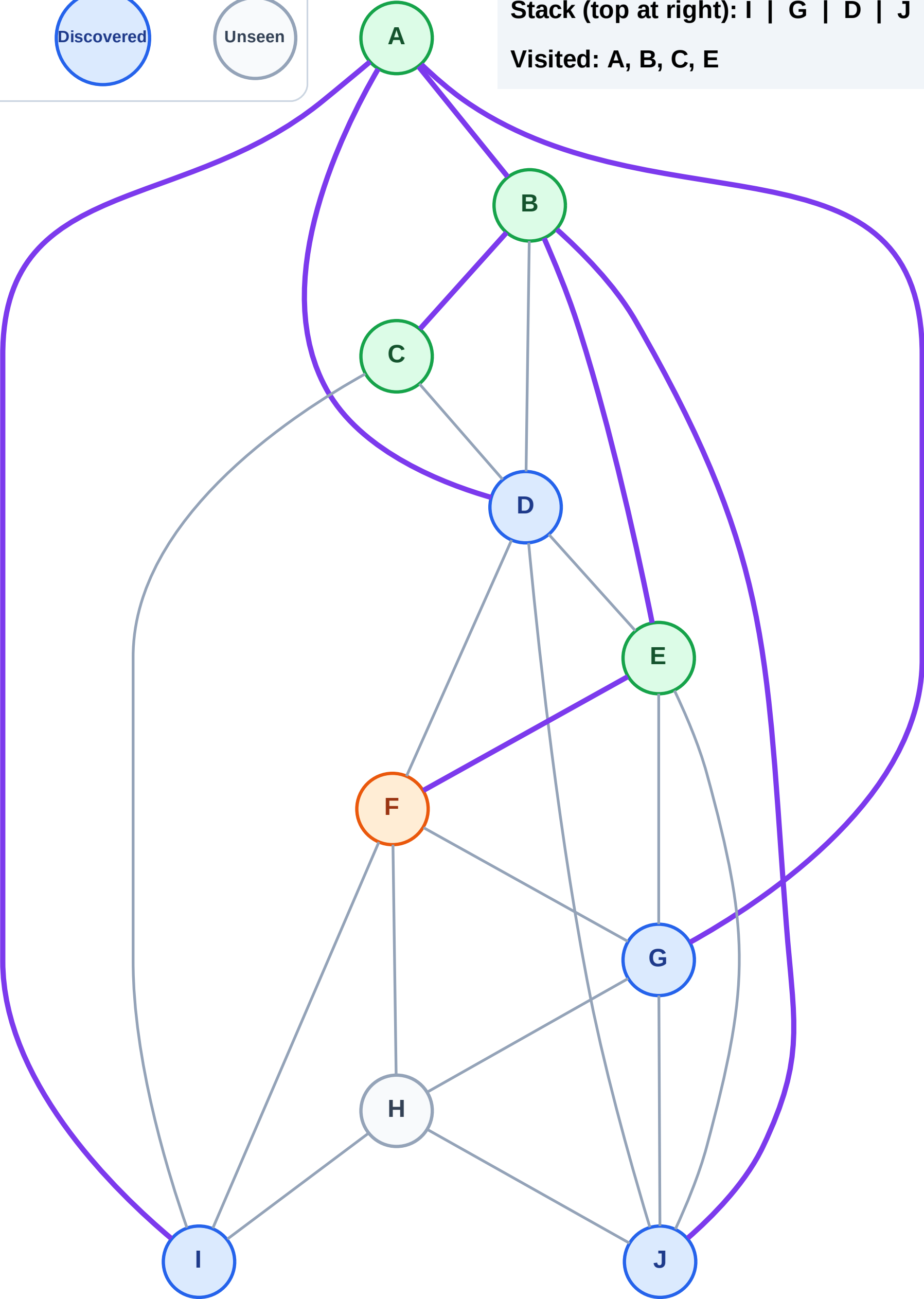
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J

Visited: A, B, C, E



DFS Traversal

Step 11: Discover H from F

Legend

Complete

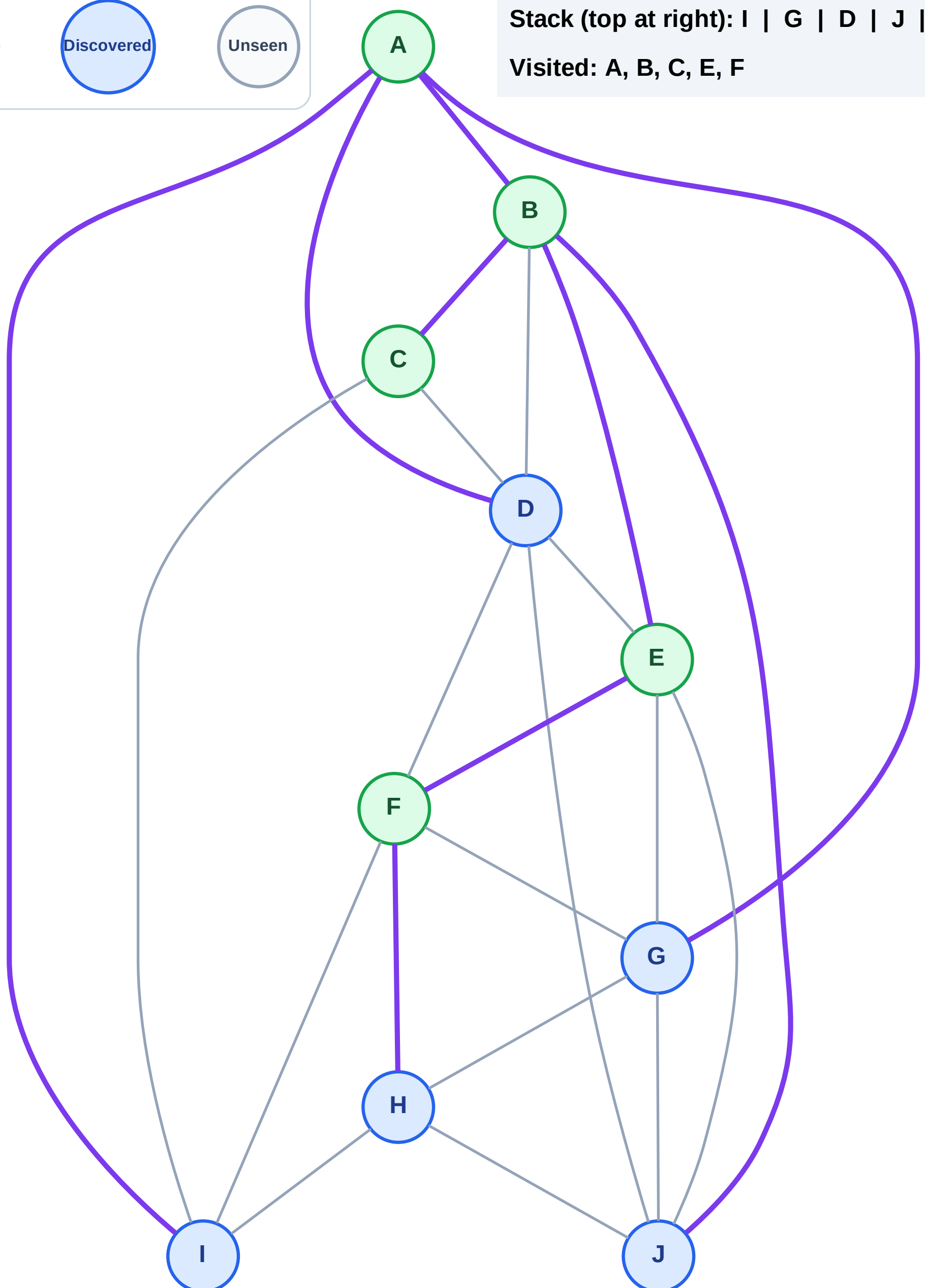
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | H

Visited: A, B, C, E, F



DFS Traversal

Step 12: Process node H

Legend

Complete

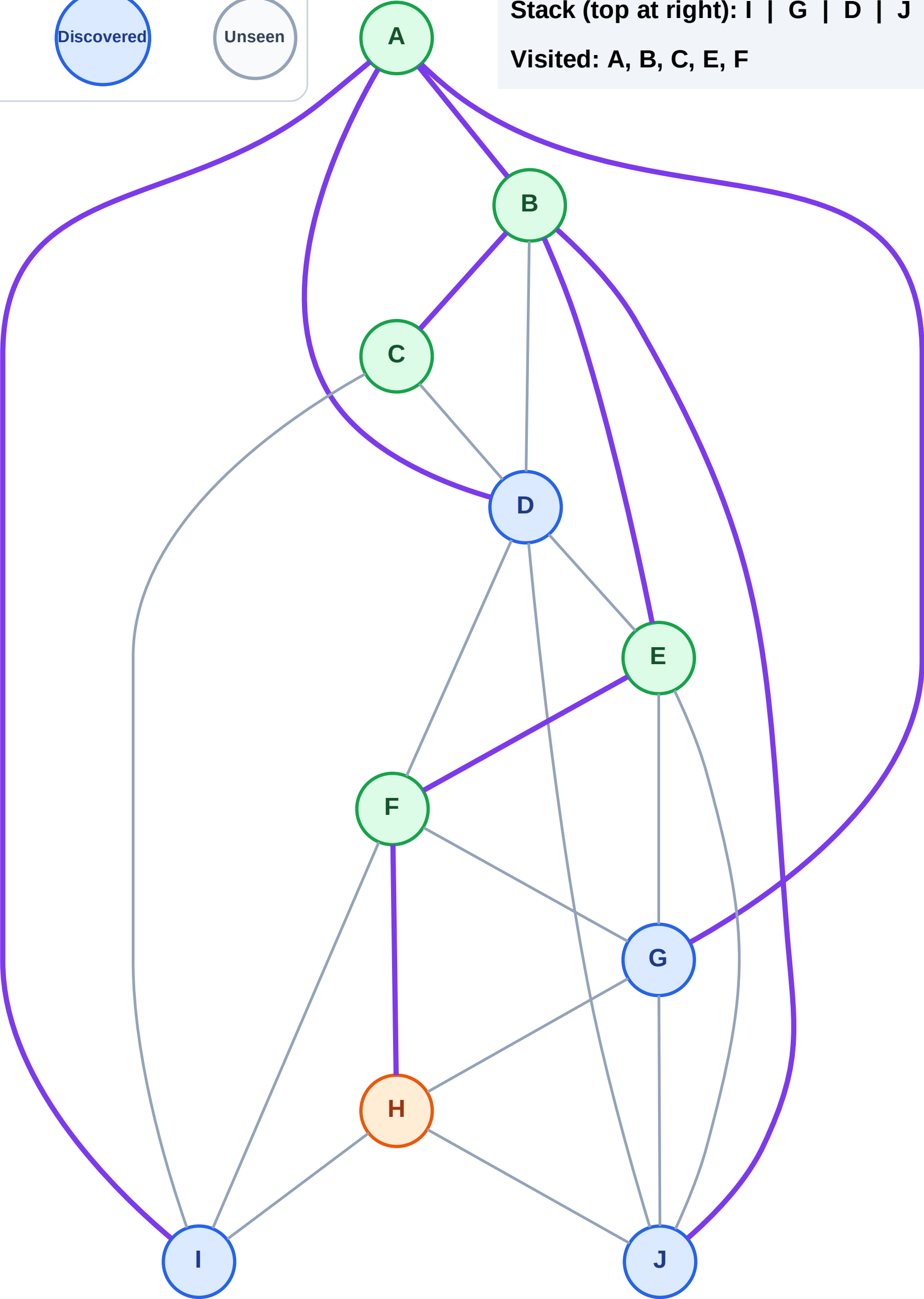
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J

Visited: A, B, C, E, F



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

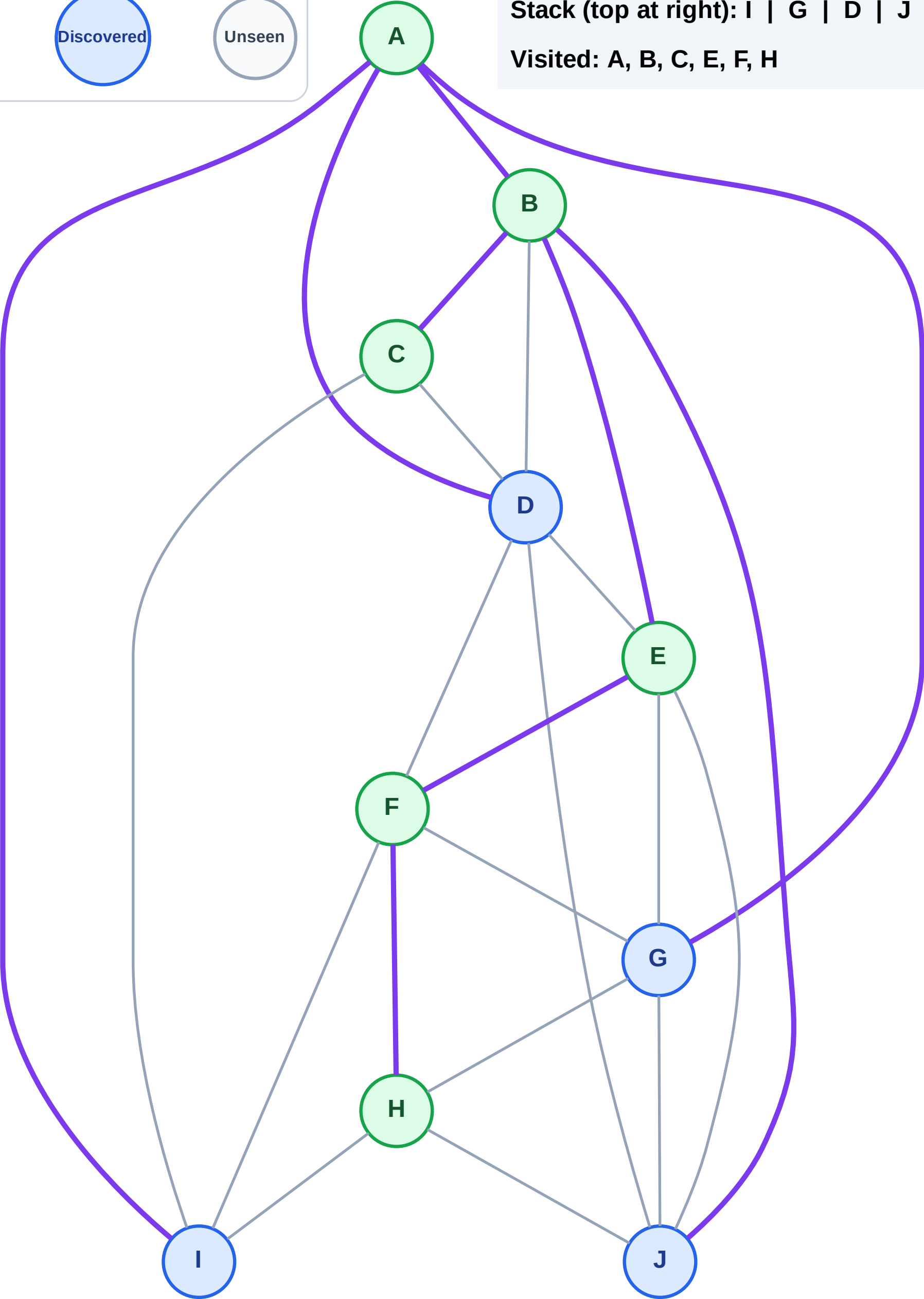
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J

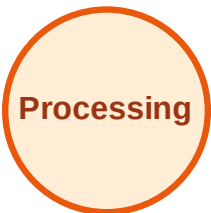
Visited: A, B, C, E, F, H



DFS Traversal

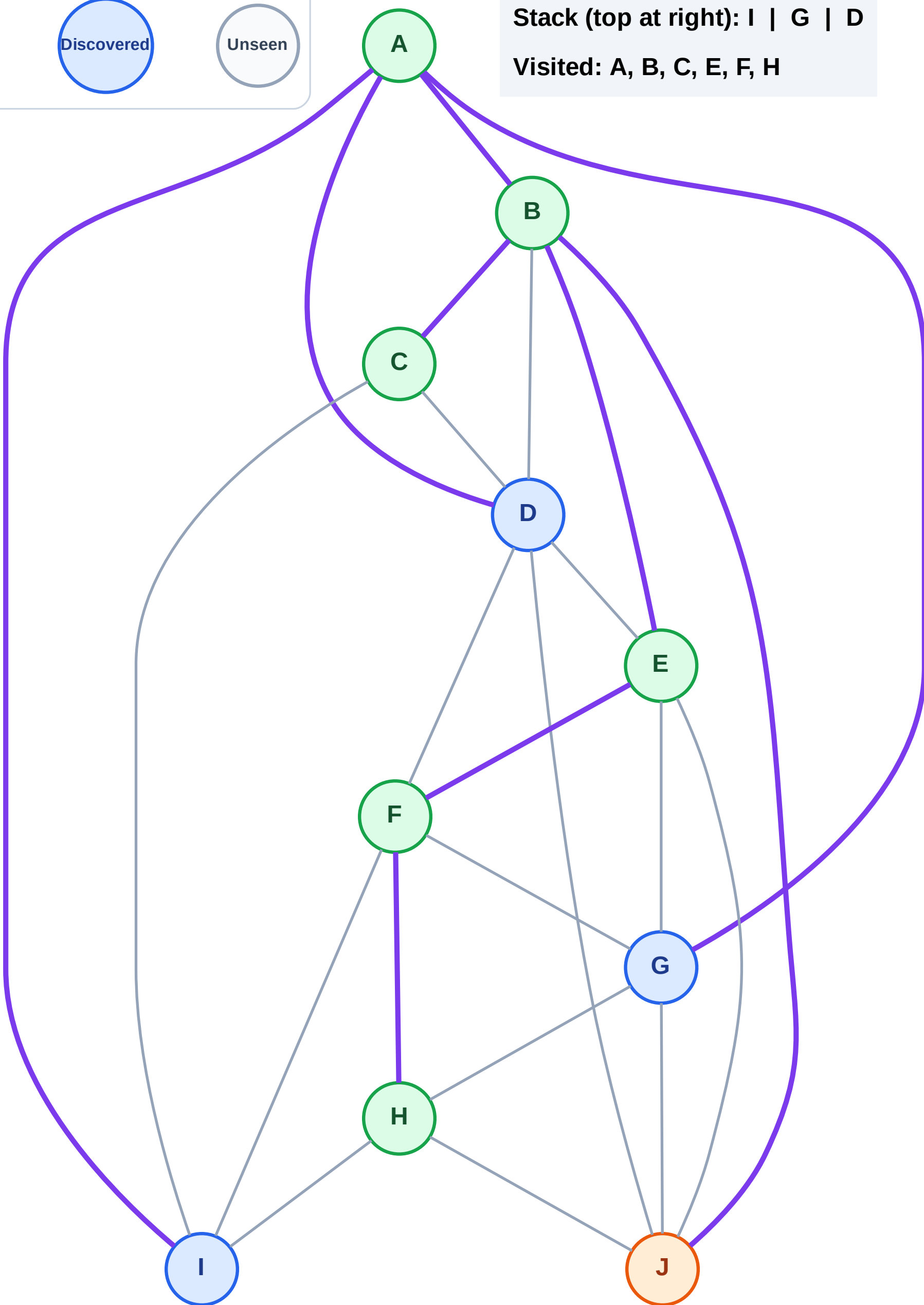
Step 14: Process node J

Legend



Stack (top at right): I | G | D

Visited: A, B, C, E, F, H



DFS Traversal

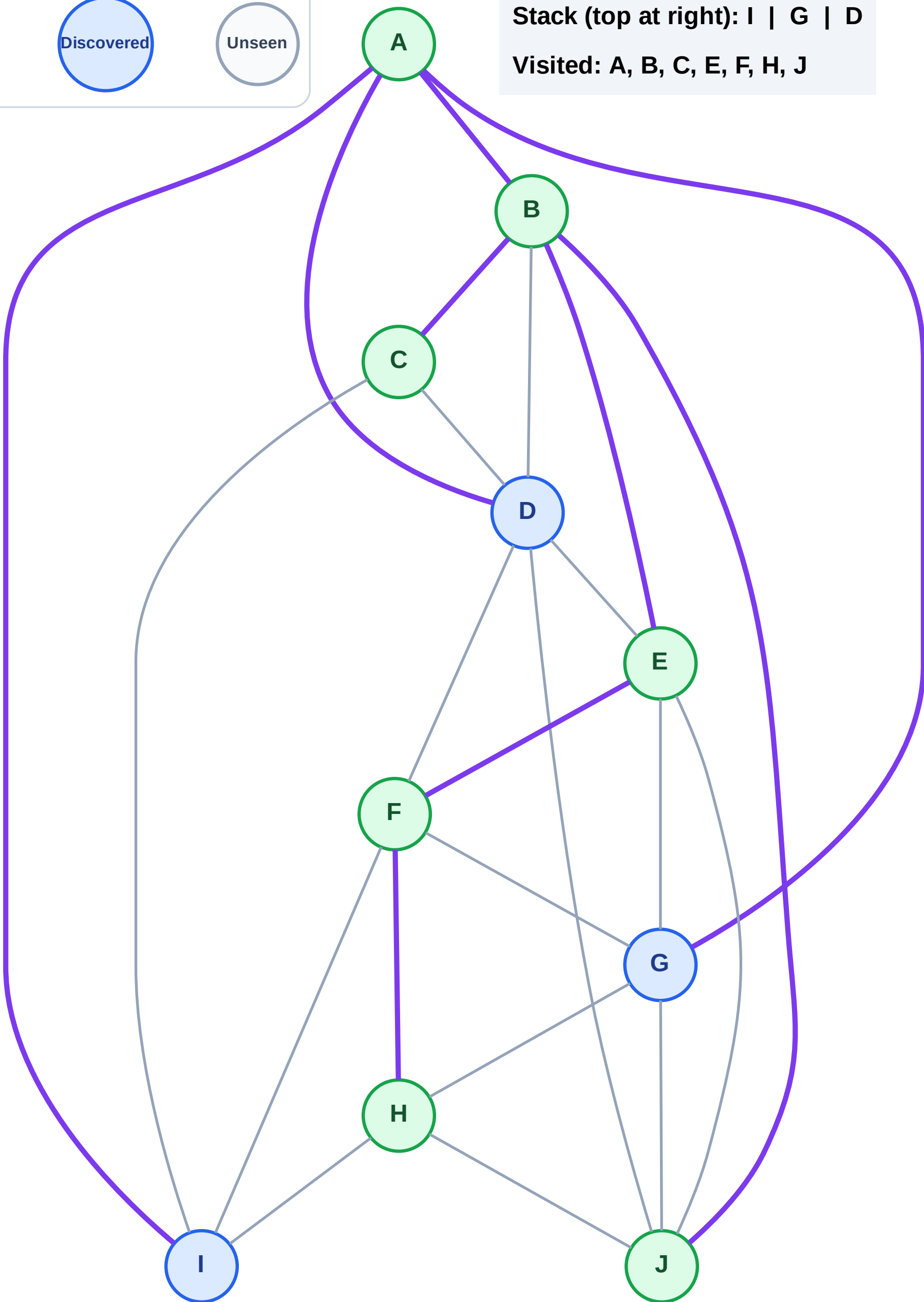
Step 15: No new neighbors from J

Legend



Stack (top at right): I | G | D

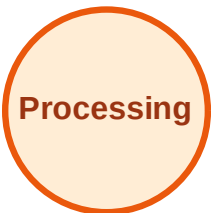
Visited: A, B, C, E, F, H, J



DFS Traversal

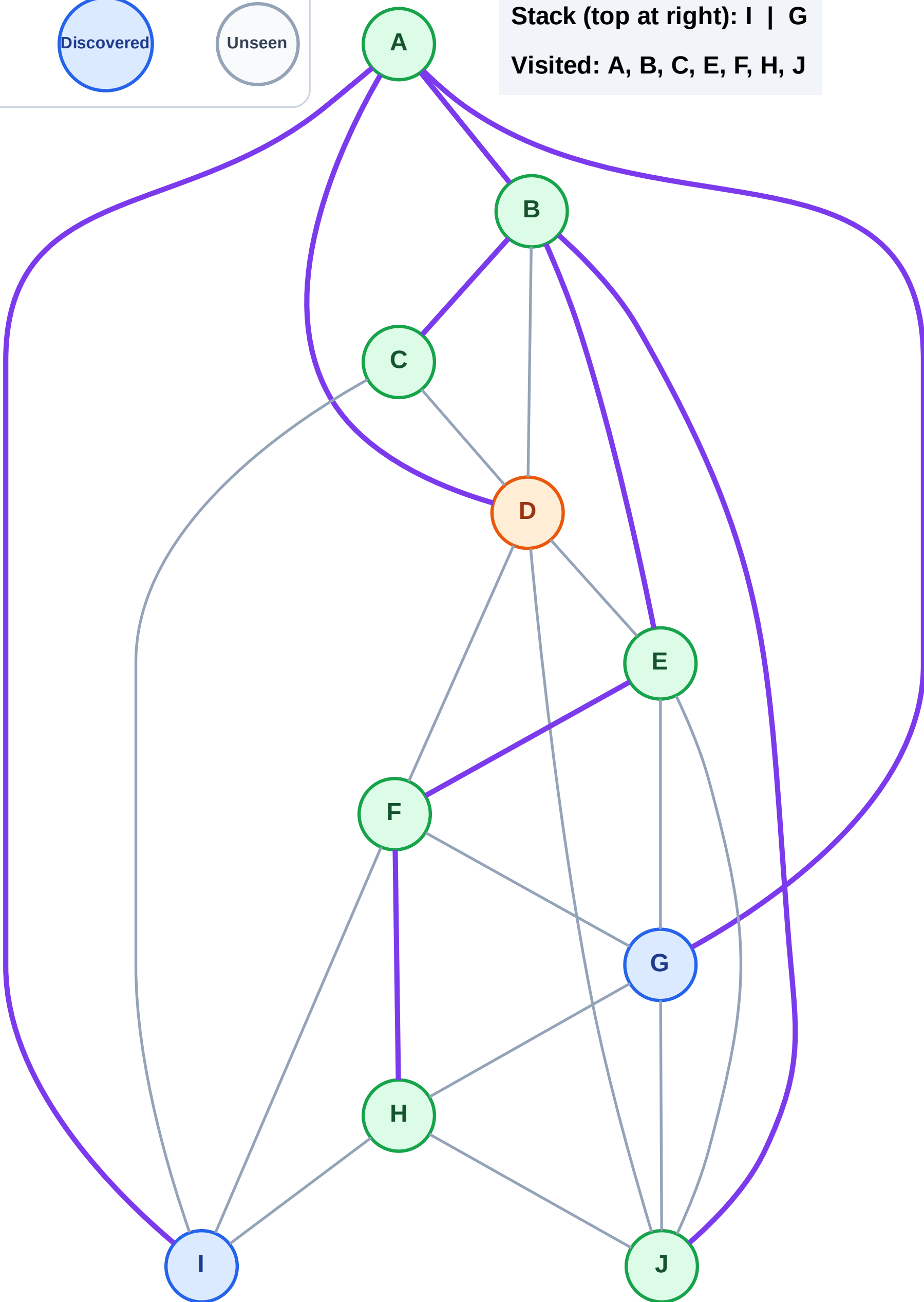
Step 16: Process node D

Legend



Stack (top at right): I | G

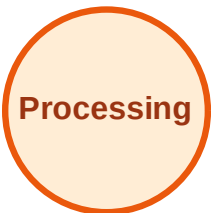
Visited: A, B, C, E, F, H, J



DFS Traversal

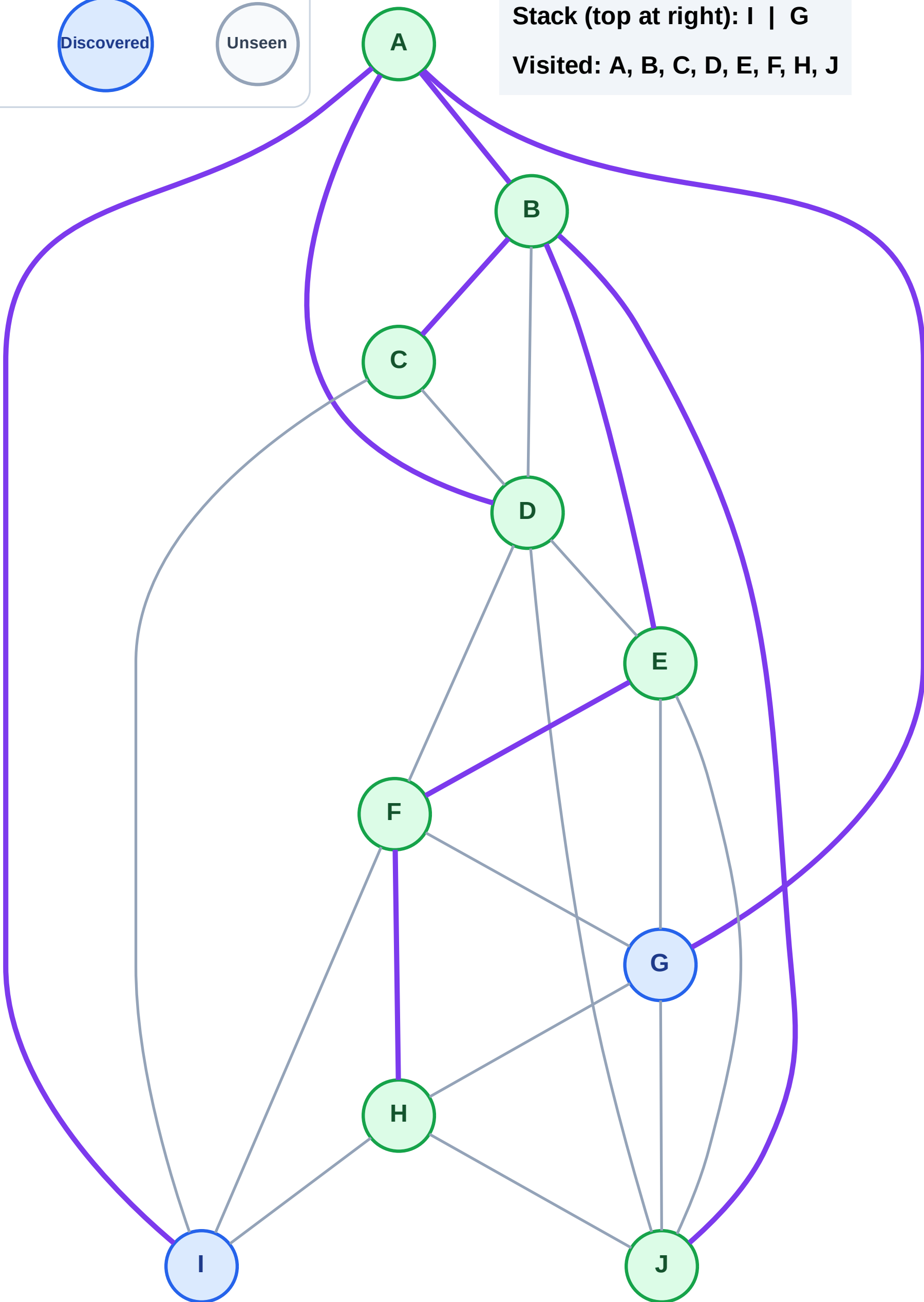
Step 17: No new neighbors from D

Legend



Stack (top at right): I | G

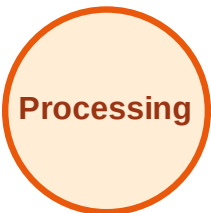
Visited: A, B, C, D, E, F, H, J



DFS Traversal

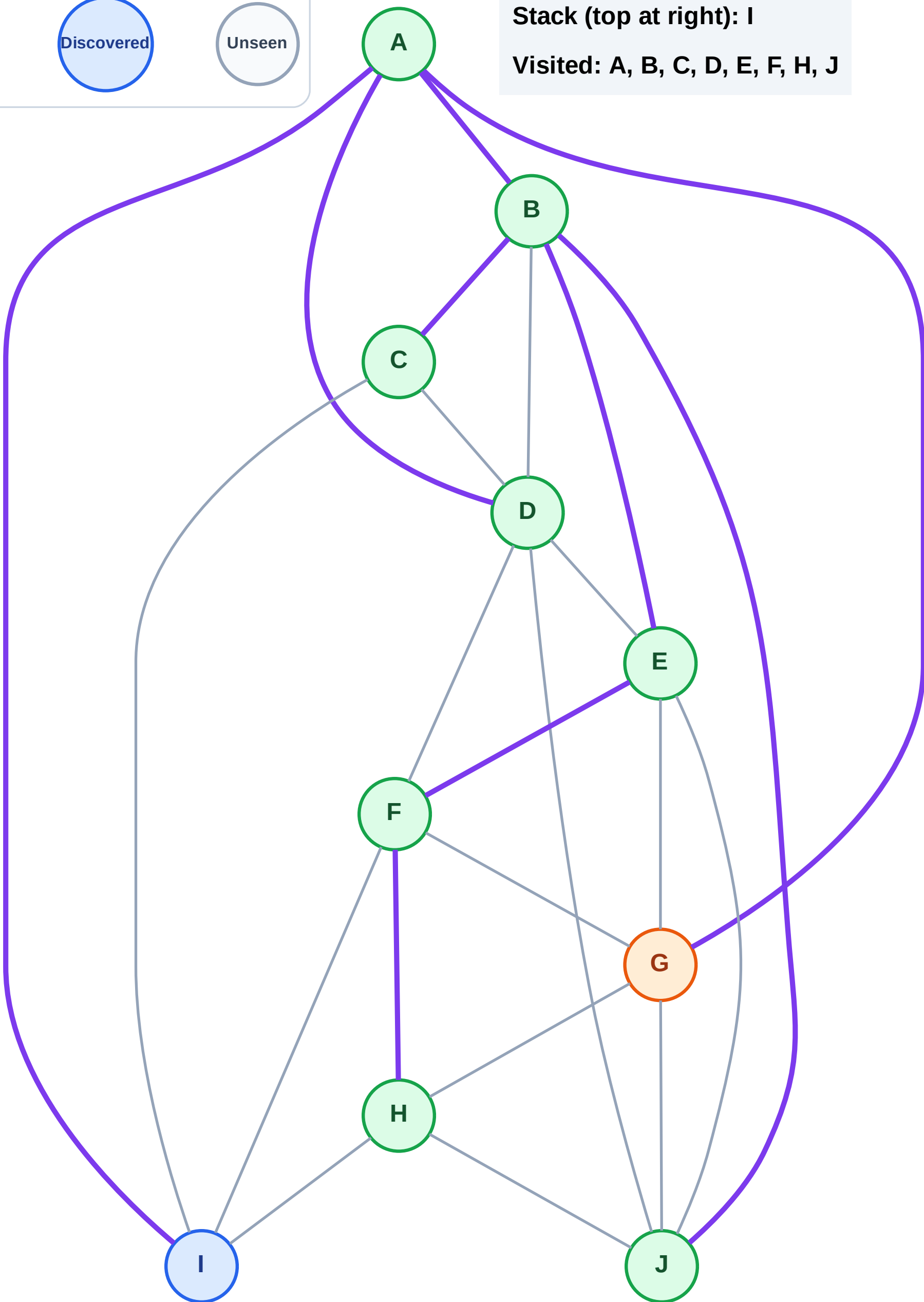
Step 18: Process node G

Legend



Stack (top at right): I

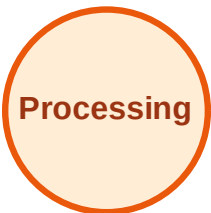
Visited: A, B, C, D, E, F, H, J



DFS Traversal

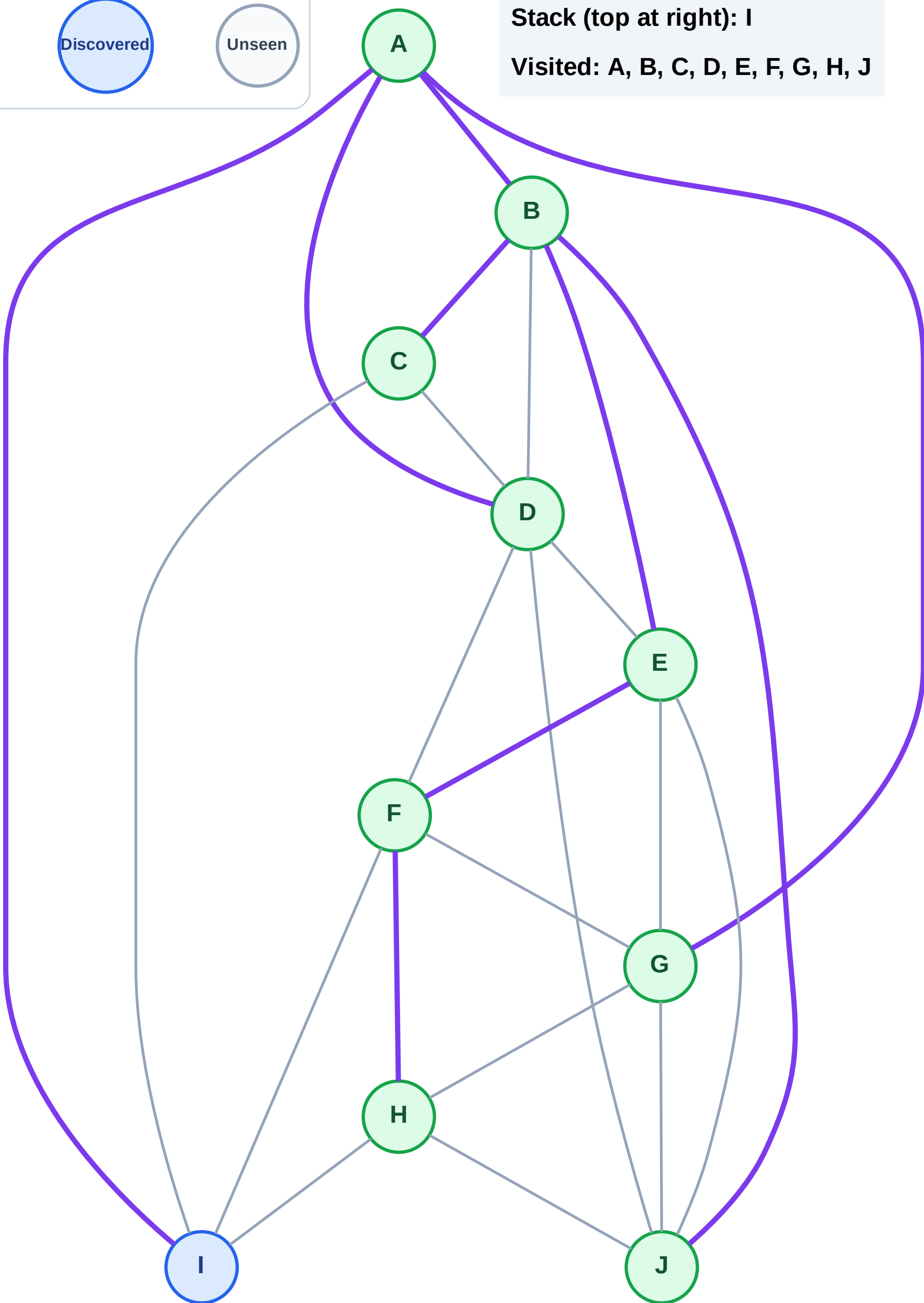
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

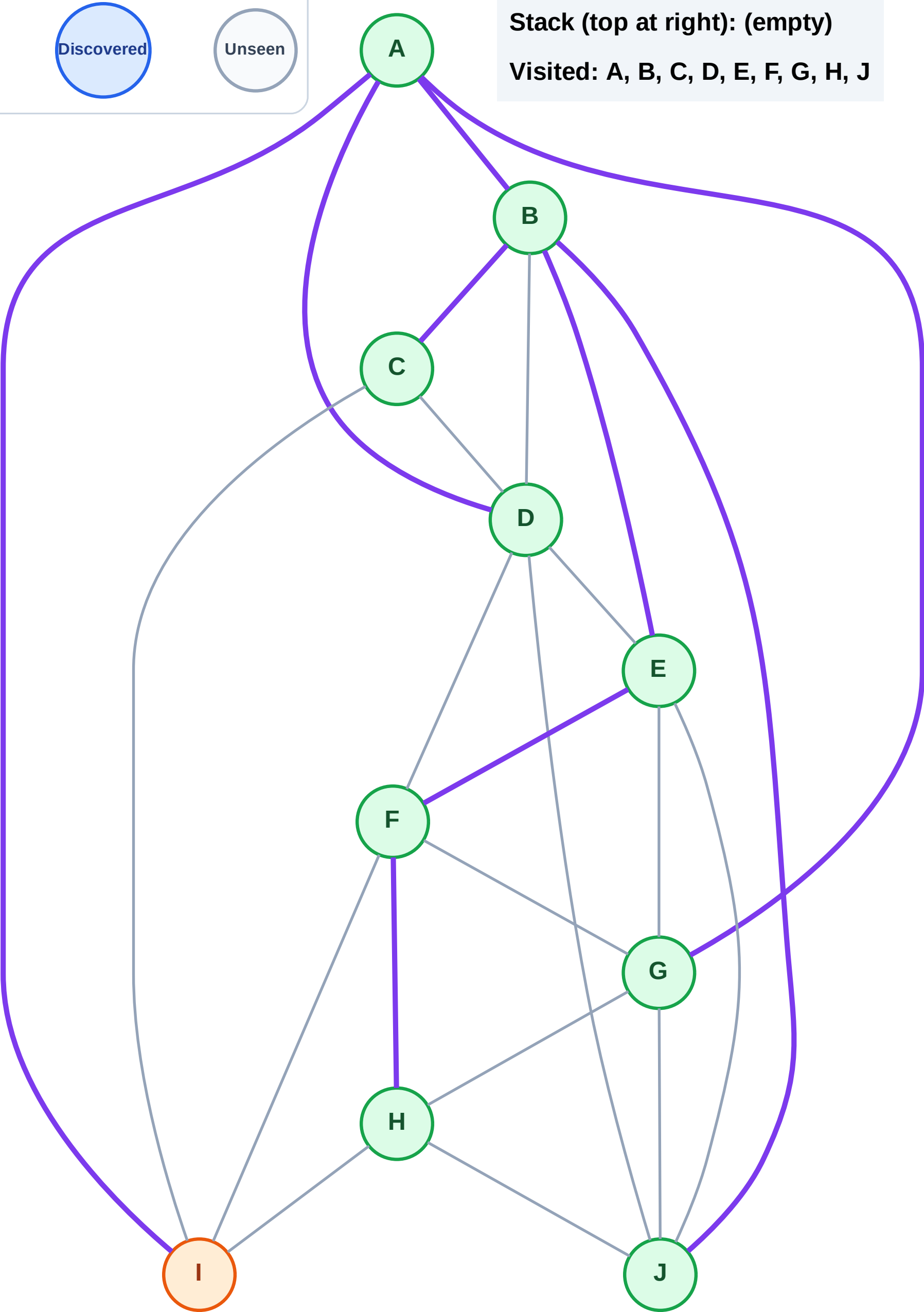
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

